

Block 3: Linear Spaces and Counting Techniques

Short explanations, worked examples and self-check questions for every unit, with every idea also explained in Malayalam.

ഓരോ ആശയവും ലളിതമായ ഇംഗ്ലീഷിലും മലയാളത്തിലും, ഉദാഹരണങ്ങളോടെ.

UNIT 8

Linear Spaces-I

ഈ unit-ൽ vector എന്നത് ഒരു ordered list of numbers ആണെന്നും, അവയെ കൂട്ടാനും scalar കൊണ്ട് ഗുണിക്കാനും, length (norm), angle (inner product) എന്നിവ കണ്ടുപിടിക്കാനും പഠിക്കുന്നു. തുടർന്ന് vector space, subspace, linear dependence, generators, basis എന്നീ ആശയങ്ങൾ $\mathbb{R}^2$, $\mathbb{R}^3$ ഉദാഹരണങ്ങളിലൂടെ മനസ്സിലാക്കാം.

Vectors: ordered n-tuples

Vector addition and scalar multiplication

Geometrical and physical interpretation

Inner (scalar / dot) product

Norm, orthogonality and angle between vectors

Vector space (linear space)

Subspaces

Linear dependence and independence

Generators (spanning sets)

Basis

Vectors: ordered n-tuples

Vector: an ordered n-tuple $(x_1, x_2, \dots, x_n)$ of real numbers; changing the order of entries gives a different vector.

Null vector: the vector $0 = (0, 0, \dots, 0)$ whose every entry is zero.

Vector: real numbers-ന്റെ ഒരു ordered n-tuple $(x_1, x_2, \dots, x_n)$; entries-ന്റെ ക്രമം മാറ്റിയാൽ മറ്റൊരു vector ആകും.

Null vector: എല്ലാ entries-ഉം പൂജ്യമായ vector $0 = (0, 0, \dots, 0)$.

In a set, order does not matter: $\{a, b, c\} = \{a, c, b\}$. A **vector** is an ordered set of numbers, so (a, b) and (b, a) are different vectors. In this unit a vector is a finite n -tuple $(x_1, x_2, \dots, x_n)$ of real numbers, called a vector over the field $\mathbb{R}$. A vector all of whose entries are zero is the **null (zero) vector** $0 = (0, 0, \dots, 0)$.

മലയാളത്തിൽ

Set-ൽ elements ഏത് order-ൽ എഴുതിയാലും ഒന്നുതന്നെ. പക്ഷേ vector-ൽ order വളരെ പ്രധാനം: $(2, 5)$ എന്നതും $(5, 2)$ എന്നതും വ്യത്യസ്തം. ഒരു ബസ് ടിക്കറ്റിലെ "seat number 2, row 5" പോലെ ആലോചിക്കുക: row-ഉം seat-ഉം മാറ്റിയാൽ വേറെ സീറ്റ് ആകും. എല്ലാ entry-യും 0 ആയ vector ആണ് null vector.

Vector over $\mathbb{R}$: $a = (a_1, a_2, \dots, a_n)$, each $a_i \in \mathbb{R}$

Row vector: (a, b, c) Column vector: the same numbers written one below another

Null vector: $0 = (0, 0, \dots, 0)$

EXAMPLE

Write the vector with entries 1, 3, -2 as a row and as a column vector. Is $(1, 3, -2) = (3, 1, -2)$?

Row: $(1, 3, -2)$. Column: $\begin{pmatrix} 1 \\ 3 \\ -2 \end{pmatrix}$

$(1, 3, -2) \neq (3, 1, -2)$, because the first entries differ ($1 \neq 3$). As sets $\{1, 3, -2\}$ and $\{3, 1, -2\}$ would be equal, but as vectors they are not.

Why it matters in AI/ML: every data point is stored as a feature vector, e.g. a house = (area, bedrooms, age) = (1200, 3, 10); swapping the order would change its meaning, which is exactly why vectors are ordered.

Vector addition and scalar multiplication

Vector addition: for vectors with the same number of entries, $x + y = (x_1 + y_1, \dots, x_n + y_n)$, adding corresponding entries.

Scalar multiplication: for a real number λ , $\lambda x = (\lambda x_1, \dots, \lambda x_n)$, multiplying every entry by λ .

Vector addition: ഒരേ എണ്ണം entries ഉള്ള vectors-ന്, $x + y = (x_1 + y_1, \dots, x_n + y_n)$; അനുബന്ധ entries കൂട്ടുന്നു.

Scalar multiplication: ഒരു real number λ -ക്ക്, $\lambda x = (\lambda x_1, \dots, \lambda x_n)$; എല്ലാ entry-യെയും λ കൊണ്ട് ഗുണിക്കുന്നു.

Two vectors can be added or subtracted only if they have the same number of entries; we add or subtract entry by entry. Two vectors are equal exactly when all corresponding entries are equal. Multiplying a vector by a real number λ (a **scalar**) multiplies every entry by λ .

മലയാളത്തിൽ

Vector കൂട്ടുമ്പോൾ ഒന്നാമത്തെ entry ഒന്നാമത്തേതിനോട്, രണ്ടാമത്തേത് രണ്ടാമത്തേതിനോട് എന്ന രീതിയിൽ മാത്രം കൂട്ടുക. രണ്ട് കടകളിലെ (അരി, പഞ്ചസാര, എണ്ണ) stock കൂട്ടുന്നത് പോലെ: അരി അരിയോട്, എണ്ണ എണ്ണയോട്. Entries-ന്റെ എണ്ണം വ്യത്യസ്തമാണെങ്കിൽ addition define ചെയ്യാൻ പറ്റില്ല. Scalar multiplication-ൽ എല്ലാ entry-യും ഒരേ number കൊണ്ട് ഗുണിക്കുന്നു, എല്ലാ സാധനങ്ങളുടെയും stock ഒരുമിച്ച് മൂന്നിരട്ടി ആക്കുന്നത് പോലെ.

$$a = (a_1, \dots, a_n), b = (b_1, \dots, b_n)$$

$$a + b = (a_1 + b_1, a_2 + b_2, \dots, a_n + b_n)$$

$$a - b = (a_1 - b_1, a_2 - b_2, \dots, a_n - b_n)$$

$$\lambda a = (\lambda a_1, \lambda a_2, \dots, \lambda a_n), \lambda \in \mathbb{R}$$

$$a = b \Leftrightarrow a_1 = b_1, a_2 = b_2, \dots, a_n = b_n$$

EXAMPLE 1 (FROM THE UNIT)

$a = (1, 3, -2)$, $b = (4, 0, 6)$. Find $a + b$, $a - b$ and $3a$.

$$a + b = (1 + 4, 3 + 0, -2 + 6) = (5, 3, 4)$$

$$a - b = (1 - 4, 3 - 0, -2 - 6) = (-3, 3, -8)$$

$$3a = (3 \cdot 1, 3 \cdot 3, 3 \cdot (-2)) = (3, 9, -6)$$

EXAMPLE 2

If $a_1 = (2, 3, 4, 7)$, $a_2 = (0, 0, 0, 1)$, $a_3 = (1, 0, 1, 0)$, find $a_1 + 2a_2 + 3a_3$.

Step 1: $2a_2 = (0, 0, 0, 2)$; $3a_3 = (3, 0, 3, 0)$.

Step 2: add entry-wise: $(2 + 0 + 3, 3 + 0 + 0, 4 + 0 + 3, 7 + 2 + 0) = (5, 3, 7, 9)$.

EXAMPLE 3

Find x and y if $(x + y, 3, x - y) = (7, 3, 1)$.

Equal vectors $\Rightarrow$ equal entries: $x + y = 7$ and $x - y = 1$. Adding, $2x = 8$, so $x = 4$, $y = 3$.

Exam tip: $(1, 2) + (1, 2, 3)$ is not defined. Always check both vectors have the same number of entries before adding.

Geometrical and physical interpretation

Definition: Geometrically, a vector is a directed line segment (arrow) having magnitude (its length) and direction; drawn from the origin, it is fixed by its end point (a_1, a_2, a_3) .

Geometric ആയി, vector എന്നാൽ magnitude-ഉം (നീളം) direction-ഉം ഉള്ള ഒരു directed line segment (arrow); origin-ൽ നിന്ന് വരച്ചാൽ അതിന്റെ end point (a_1, a_2, a_3) കൊണ്ട് അത് പൂർണ്ണമായി നിശ്ചയിക്കപ്പെടുന്നു.

Geometrically a vector is an arrow: its length is the **magnitude** and the arrowhead shows the **direction**. Physical quantities like force depend only on magnitude and direction, not on where the arrow is placed, so we always draw vectors starting at the origin. Then the vector is completely fixed by the point where its head ends: there is a one-to-one correspondence between points (a_1, a_2, a_3) and vectors from the origin, which is why we write $a = (a_1, a_2, a_3)$. In a plane a vector needs 2 numbers (magnitude + one angle); in space it needs 3 (magnitude + two angles θ, ϕ).

മലയാളത്തിൽ

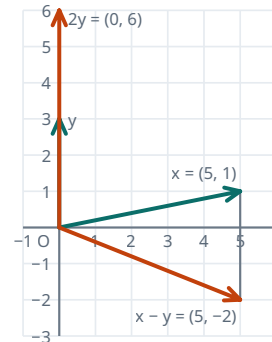
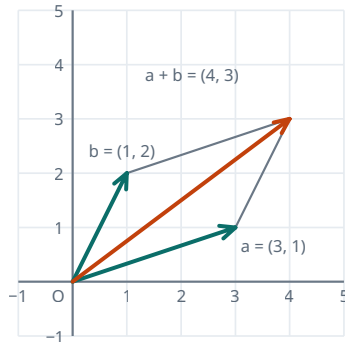
Vector-നെ ഒരു അമ്പ് (arrow) ആയി വരയ്ക്കാം: നീളം magnitude, അമ്പിന്റെ മുന്ന direction. എല്ലാ arrow-ഉം origin-ൽ നിന്ന് തുടങ്ങുന്നു എന്ന് തീരുമാനിച്ചാൽ, arrow-ന്റെ മുന്ന എവിടെ എത്തുന്നു എന്ന point മാത്രം പറഞ്ഞാൽ മതി. ഉദാഹരണത്തിന് $(3, 1)$ എന്നാൽ "വലത്തോട്ട് 3, മുകളിലേക്ക് 1". Addition ജ്യോമിതീയമായി parallelogram law ആണ്: a -യുടെ മുന്നയിൽ നിന്ന് b നടന്നാൽ $a + b$ -ൽ എത്തും.

For $a = (a_1, a_2, a_3)$:

$$\text{Magnitude} = \sqrt{a_1^2 + a_2^2 + a_3^2}$$

$$\text{Direction: } \tan \theta = a_2/a_1, \quad \cos \varphi = a_3/\sqrt{a_1^2 + a_2^2 + a_3^2}$$

Parallelogram law: $a + b$ is the diagonal of the parallelogram with sides a and b



Vector addition: $(3, 1) + (1, 2) = (4, 3)$, the diagonal of the parallelogram.

$x = (5, 1)$, $y = (0, 3)$: $2y = (0, 6)$
and $x - y = (5, -2)$.

EXAMPLE 1

Given $x = (5, 1)$ and $y = (0, 3)$, find $2y$ and $x - y$ graphically.

$2y$: stretch y to twice its length in the same direction: $2y = (0, 6)$, an arrow up the y -axis to height 6.

$x - y = x + (-y)$: from the head of $x = (5, 1)$ move 3 units down (that is $-y$). You land at $(5, -2)$, so $x - y = (5, -2)$. See the right-hand figure.

EXAMPLE 2

Find the magnitude and direction angles of $a = (1, 1, \sqrt{2})$.

$$\text{Magnitude} = \sqrt{1 + 1 + 2} = \sqrt{4} = 2.$$

$$\tan \theta = a_2/a_1 = 1/1 = 1 \Rightarrow \theta = \pi/4 \text{ (45}^\circ\text{)}.$$

$$\cos \varphi = a_3/2 = \sqrt{2}/2 \Rightarrow \varphi = \pi/4 \text{ (45}^\circ\text{)}.$$

Exam tip: λa for $\lambda > 0$ points the same way as a and is λ times as long; for $\lambda < 0$ it points the opposite way. $-a$ is a reversed arrow of the same length.

Inner (scalar / dot) product

Definition: The inner (scalar/dot) product of $x = (x_1, \dots, x_n)$ and $y = (y_1, \dots, y_n)$ is the real number $\langle x, y \rangle = x_1y_1 + x_2y_2 + \dots + x_ny_n$.

$x = (x_1, \dots, x_n)$, $y = (y_1, \dots, y_n)$ എന്നിവയുടെ inner (scalar/dot) product എന്നത് $\langle x, y \rangle = x_1y_1 + x_2y_2 + \dots + x_ny_n$ എന്ന real number ആണ്.

Addition and scalar multiplication give back a vector. The **inner product** takes two vectors of the same length and gives a single real number: multiply corresponding entries and add. It is also called the **scalar product** or **dot product**. Do not confuse it with scalar multiplication (number $\times$ vector).

മലയാളത്തിൽ

Inner product-ന്റെ ഉത്തരം ഒരു vector അല്ല, ഒരു number ആണ്. ഒരു ബില്ലിന് പോലെ ആലോചിക്കുക: quantities (2 kg, 1 kg, 5 kg), prices (₹3, ₹2, ₹1) എന്നിവ ജോടിയിൽ ഗുണിച്ച് കൂട്ടിയാൽ total bill = $2 \cdot 3 + 1 \cdot 2 + 5 \cdot 1 = ₹13$. ഇതാണ് dot product. IGNOU ഇത് $\langle a, b \rangle$ എന്ന് എഴുതുന്നു; ഇവിടെ $\langle a, b \rangle$ എന്നും കാണാം.

$$\langle a, b \rangle = a_1b_1 + a_2b_2 + \dots + a_nb_n = \sum_{i=1}^n a_ib_i$$

Also written $\langle a, b \rangle$ or $a \cdot b$

$$\langle a, b \rangle = \langle b, a \rangle; \quad \langle a, a \rangle = a_1^2 + \dots + a_n^2 \geq 0$$

EXAMPLE 1 (FROM THE UNIT)

$a = (1, -1, 2)$, $b = (-2, 3, 6)$. Find $\langle a, b \rangle$.

$$\langle a, b \rangle = (1)(-2) + (-1)(3) + (2)(6) = -2 - 3 + 12 = 7.$$

EXAMPLE 2 (FROM THE UNIT)

(i) $a = (2, -1, 5)$, $b = (3, 2, 1)$ (ii) $a = (1, -1, 2)$, $b = (1, -1, -2)$

$$(i) (2)(3) + (-1)(2) + (5)(1) = 6 - 2 + 5 = 9$$

$$(ii) (1)(1) + (-1)(-1) + (2)(-2) = 1 + 1 - 4 = -2$$

EXAMPLE 3

Find $\langle a, b \rangle$ for (i) $(2, 3, 4)$, $(4, 5, 5)$ and (ii) $(-2, -3, 4)$, $(4, 5, -6)$.

$$(i) 8 + 15 + 20 = 43$$

$$(ii) -8 - 15 - 24 = -47$$

Exam tip: watch signs: $(-3)(5) = -15$ and $(4)(-6) = -24$. Most lost marks here are sign slips. The answer is always a number, never a vector.

Norm, orthogonality and angle between vectors

Norm: $\|x\| = \sqrt{\langle x, x \rangle} = \sqrt{(x_1^2 + \dots + x_n^2)}$, the length of x .

Orthogonal: x and y are orthogonal if $\langle x, y \rangle = 0$.

Angle: for non-zero a, b , the angle θ is given by $\cos \theta = \langle a, b \rangle / (\|a\| \cdot \|b\|)$.

Norm: $\|x\| = \sqrt{\langle x, x \rangle} = \sqrt{(x_1^2 + \dots + x_n^2)}$; x -ന്റെ നീളം.

Orthogonal: $\langle x, y \rangle = 0$ ആണെങ്കിൽ x -ഉം y -ഉം orthogonal ആണ്.

Angle: പൂജ്യമല്ലാത്ത a, b -ക്ക്, $\cos \theta = \langle a, b \rangle / (\|a\| \cdot \|b\|)$ ആയ θ ആണ് അവയ്ക്കിടയിലുള്ള angle.

The **norm** (length, magnitude) of x is the positive square root of the inner product of x with itself. Two vectors are **orthogonal** (perpendicular) if their inner product is 0; the null vector is orthogonal to every vector. The angle θ between two non-zero vectors is given by $\cos \theta = \langle a, b \rangle / (\|a\| \cdot \|b\|)$. Schwarz's inequality guarantees this value lies between -1 and 1 .

മലയാളത്തിൽ

Norm എന്നാൽ vector-ന്റെ നീളം: entries-ന്റെ squares കൂട്ടി square root എടുക്കുക, Pythagoras theorem തന്നെ. $(3, 4)$ -ന്റെ നീളം 5, ഒരു വലത് കോൺ ത്രികോണത്തിലെ hypotenuse പോലെ. Dot product 0 ആയാൽ രണ്ട് vectors-ഉം 90° ൽ നിൽക്കുന്നു, അതാണ് orthogonal. മറ്റ് angles-ന് $\cos \theta$ formula ഉപയോഗിക്കുക.

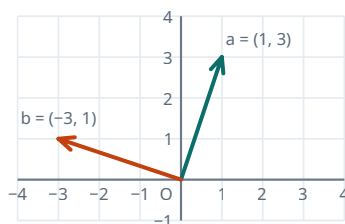
$$\|x\| = \ell(x) = \sqrt{\langle x, x \rangle} = \sqrt{(x_1^2 + x_2^2 + \dots + x_n^2)} = [\sum x_i^2]^{1/2}$$

$$\text{Orthogonal: } \langle a, b \rangle = 0$$

$$\cos \theta = \langle a, b \rangle / (\|a\| \cdot \|b\|)$$

$$\text{Schwarz's inequality: } (\sum a_i b_i)^2 \leq (\sum a_i^2)(\sum b_i^2) \Rightarrow \cos^2 \theta \leq 1$$

$$\text{If } b = \lambda a: \|b\| = |\lambda| \|a\|, \langle a, b \rangle = \lambda \|a\|^2, \cos \theta = \pm 1 \quad (\theta = 0 \text{ if } \lambda > 0, \theta = \pi \text{ if } \lambda < 0)$$



$a = (1, 3)$ and $b = (-3, 1)$: $\langle a, b \rangle = -3 + 3 = 0$, so they are perpendicular.

EXAMPLE 1

Find $\|a\|$ for $a = (5, -2, 3)$, and the norms of $(2, 5)$ and $(-2, 2)$.

$\langle a, a \rangle = 25 + 4 + 9 = 38$, so $\|a\| = \sqrt{38}$ (not 38).

$\|(2, 5)\| = \sqrt{4 + 25} = \sqrt{29}$; $\|(-2, 2)\| = \sqrt{4 + 4} = \sqrt{8} = 2\sqrt{2}$.

EXAMPLE 2 (FROM THE UNIT)

Find the angle between (i) $a = (2, -1, 5)$, $b = (2, -1, -1)$ and (ii) $a = (3, 1, 2)$, $b = (1, 3, -3)$.

(i) $\langle a, b \rangle = 4 + 1 - 5 = 0 \Rightarrow \cos \theta = 0 \Rightarrow \theta = \pi/2$.

(ii) $\langle a, b \rangle = 3 + 3 - 6 = 0 \Rightarrow \cos \theta = 0 \Rightarrow \theta = \pi/2$. Both pairs are orthogonal.

EXAMPLE 3

Find the angle between $a = (1, 0)$ and $b = (1, 1)$.

$\langle a, b \rangle = 1 \cdot 1 + 0 \cdot 1 = 1$; $\|a\| = 1$; $\|b\| = \sqrt{2}$.

$\cos \theta = 1/(1 \cdot \sqrt{2}) = 1/\sqrt{2} \Rightarrow \theta = \pi/4$ (45°).

Exam tip: the unit's worked example writes $\|a\| = 38$ for $a = (5, -2, 3)$; 38 is $\|a\|^2$, the norm is $\sqrt{38}$ (the unit itself says $\sqrt{38}$ a few lines later). Likewise, the answer key gives the norms of $(2, 5)$ and $(-2, 2)$ as 9 and 8; the correct norms are $\sqrt{29}$ and $\sqrt{8} = 2\sqrt{2}$. Also, for $b = \lambda a$ the unit writes $\langle a, b \rangle = \lambda \|a\|$ and $\theta = 0$; correctly $\langle a, b \rangle = \lambda \|a\|^2$, and $\theta = 0$ only when $\lambda > 0$ ($\theta = \pi$ when $\lambda < 0$).

Vector space (linear space)

Definition: A vector (linear) space over $\mathbb{R}$ is a non-empty set V with addition and scalar multiplication under which V is closed and which satisfy the vector space axioms (commutativity, associativity, distributivity, zero vector, negatives).

$\mathbb{R}$ -ന് മുകളിലുള്ള vector (linear) space എന്നാൽ addition, scalar multiplication എന്നിവയ്ക്ക് കീഴിൽ closed ആയതും vector space axioms (commutativity, associativity, distributivity, zero vector, negatives) പാലിക്കുന്നതുമായ ഒരു non-empty set V .

Let $V = \{(x_1, x_2, \dots, x_n) \mid x_i \in \mathbb{R}\}$, the set of all n -component real vectors. With entry-wise addition and scalar multiplication, V satisfies the properties below for all vectors u, v, w and scalars α, β . Any set with two operations satisfying all these properties is called a **vector space** or **linear space** over $\mathbb{R}$. In short: V is closed under addition and scalar multiplication, addition is commutative and associative, scalar multiplication distributes over addition, and V contains the null vector and negatives.

മലയാളത്തിൽ

Vector space എന്നാൽ ഒരു "അടച്ച മുറി" പോലെ: അകത്തുള്ള ഏത് രണ്ട് vectors കൂട്ടിയാലും, ഏത് vector-നെ ഏത് real number കൊണ്ട് ഗുണിച്ചാലും ഫലം മുറിക്കുള്ളിൽ തന്നെ നിൽക്കണം (closure). കൂടാതെ സാധാരണ സംഖ്യകളിൽ നമ്മൾ ഉപയോഗിക്കുന്ന നിയമങ്ങൾ ($u + v = v + u$ തുടങ്ങിയവ) പാലിക്കണം. $\mathbb{R}^2, \mathbb{R}^3$ എന്നിവ ഏറ്റവും നല്ല ഉദാഹരണങ്ങൾ.

For $u, v, w \in V$ and $\alpha, \beta \in \mathbb{R}$:

1. $u + v \in V$ (closed under addition)
 2. $u + v = v + u$
 3. $(u + v) + w = u + (v + w)$
 4. $u + 0 = 0 + u = u$, where $0 = (0, 0, \dots, 0)$
 5. For each u there is $-u$ with $u + (-u) = (-u) + u = 0$
 6. $(\alpha + \beta)u = \alpha u + \beta u$
 7. $\alpha(u + v) = \alpha u + \alpha v$
 8. $(\alpha\beta)u = \alpha(\beta u)$
 9. $1u = u$
- (also) $\alpha u \in V$ (closed under scalar multiplication)

EXAMPLE 1: CHECKING THE AXIOMS IN $\mathbb{R}^2$ WITH NUMBERS

Take $u = (1, 2)$, $v = (3, -1)$, $w = (0, 4)$, $\alpha = 2$, $\beta = 3$.

Closure: $u + v = (4, 1) \in \mathbb{R}^2$; $2u = (2, 4) \in \mathbb{R}^2$. ✓

Commutative: $u + v = (4, 1)$; $v + u = (3 + 1, -1 + 2) = (4, 1)$. ✓

Associative: $(u + v) + w = (4, 1) + (0, 4) = (4, 5)$; $u + (v + w) = (1, 2) + (3, 3) = (4, 5)$. ✓

Zero and negative: $u + 0 = (1, 2)$; $-u = (-1, -2)$ and $u + (-u) = (0, 0)$. ✓

$(\alpha + \beta)u = 5(1, 2) = (5, 10)$; $\alpha u + \beta u = (2, 4) + (3, 6) = (5, 10)$. ✓

$\alpha(u + v) = 2(4, 1) = (8, 2)$; $\alpha u + \alpha v = (2, 4) + (6, -2) = (8, 2)$. ✓

$(\alpha\beta)u = 6(1, 2) = (6, 12)$; $\alpha(\beta u) = 2(3, 6) = (6, 12)$. ✓ $1u = (1, 2) = u$. ✓

In general each check reduces to the same law for real numbers in each entry, e.g. $(u + v)_i = u_i + v_i = v_i + u_i$.

So $\mathbb{R}^n$ is a vector space.

EXAMPLE 2: A SET THAT IS NOT A VECTOR SPACE

Is $V = \{(x, y) \mid x \geq 0, y \geq 0\}$ (the first quadrant) a vector space under the usual operations?

Addition stays inside: $(1, 2) + (3, 4) = (4, 6)$ ✓. But scalar multiplication fails: $(-1)(1, 2) = (-1, -2)$, which has negative entries, so it is not in V . Also $(1, 2)$ has no negative inside V . Hence V is **not** a vector space.

Exam tip: the unit prints axiom 3 as $(u + v) + w = u + (u + w)$ (a typo; it should be $u + (v + w)$), says "satisfying i, ii and iii" (all the listed properties are needed), and omits closure under scalar multiplication from the list though its summary uses it. To show a set is not a vector space, one failing axiom with a concrete counterexample is enough.

Subspaces

Definition: A non-empty subset W of a vector space V is a subspace if W is itself a vector space under the same operations, i.e. $0 \in W$ and W is closed under addition and scalar multiplication.

ഒരു vector space V -യുടെ non-empty subset W അതേ operations-ന് കീഴിൽ സ്വയം ഒരു vector space ആണെങ്കിൽ, അതായത് $0 \in W$ -ഉം W addition, scalar multiplication എന്നിവയ്ക്ക് കീഴിൽ closed-ഉം ആണെങ്കിൽ, W ഒരു subspace ആണ്.

A non-empty subset W of a vector space V is a **subspace** if W , with the same addition and scalar multiplication, is itself a vector space. In practice it is enough to check that W contains 0 and is closed under addition and scalar multiplication (the other laws are inherited from V). In the plane $V_2 = \mathbb{R}^2$, the x-axis, the y-axis, and every line through the origin are subspaces.

മലയാളത്തിൽ

Subspace എന്നാൽ വലിയ vector space-ന്റെ ഉള്ളിലുള്ള ഒരു ചെറിയ vector space. വലിയ വീടിനുള്ളിലെ ഒരു മുറി പോലെ, പക്ഷേ ആ മുറിയും സ്വന്തമായി "അടച്ചത്" ആയിരിക്കണം. $\mathbb{R}^2$ -ൽ origin വഴി പോകുന്ന ഏത് straight line-ഉം subspace ആണ്. Origin വഴി പോകാത്ത line subspace അല്ല, കാരണം അതിൽ 0 ഇല്ല.

Subspace test: $W \subseteq V$ is a subspace if

(i) $0 \in W$, (ii) $u, v \in W \Rightarrow u + v \in W$, (iii) $u \in W, \alpha \in \mathbb{R} \Rightarrow \alpha u \in W$

Subspaces of V_2 :

$W = \{(x_1, x_2) \mid x_1 \in \mathbb{R}, x_2 = 0\}$ (x-axis)

$W = \{(x_1, x_2) \mid x_2 \in \mathbb{R}, x_1 = 0\}$ (y-axis)

$W = \{(x_1, x_2) \mid c_1 x_1 + c_2 x_2 = 0\}$ (line through origin)

EXAMPLE 1

Show $W = \{(x, y) \mid y = 2x\}$ is a subspace of $\mathbb{R}^2$.

(i) $(0, 0)$: $0 = 2 \cdot 0$ ✓, so $0 \in W$.

(ii) Take $(a, 2a)$ and $(b, 2b)$ in W . Sum $= (a + b, 2a + 2b) = (a + b, 2(a + b)) \in W$ ✓.

(iii) $\alpha(a, 2a) = (\alpha a, 2(\alpha a)) \in W$ ✓. Hence W is a subspace (the line $y = 2x$ through the origin).

EXAMPLE 2

Is $W = \{(x, y) \mid y = 2x + 1\}$ a subspace of $\mathbb{R}^2$?

$(0, 0)$: $0 \neq 2 \cdot 0 + 1$, so $0 \notin W$. Not a subspace. (Also $(0, 1) + (1, 3) = (1, 4)$, but $2 \cdot 1 + 1 = 3 \neq 4$, so addition fails too.)

EXAMPLE 3 ($\mathbb{R}^3$)

Show $W = \{(x, y, z) \mid 2x + y - z = 0\}$ is a subspace of $\mathbb{R}^3$.

$0 = (0, 0, 0)$ satisfies $0 + 0 - 0 = 0$ ✓. If u, v satisfy $2x + y - z = 0$ then for $u + v$: $2(u_1 + v_1) + (u_2 + v_2) - (u_3 + v_3) = 0 + 0 = 0$ ✓. For αu : $\alpha(2u_1 + u_2 - u_3) = \alpha \cdot 0 = 0$ ✓. So W (a plane through the origin) is a subspace.

Exam tip: the quickest way to reject a subset is " $0 \notin W$ ". A set defined by equations with a non-zero constant on one side (like $x + y = 1$) is never a subspace.

Linear dependence and independence

Linearly dependent: $x_1, \dots, x_n$ are linearly dependent if $c_1x_1 + \dots + c_nx_n = 0$ for some scalars c_i not all zero.

Linearly independent: they are linearly independent if $c_1x_1 + \dots + c_nx_n = 0$ holds only when $c_1 = \dots = c_n = 0$.

Linearly dependent: എല്ലാം പൂജ്യമല്ലാത്ത ചില scalars c_i -ക്ക് $c_1x_1 + \dots + c_nx_n = 0$ ആണെങ്കിൽ $x_1, \dots, x_n$ linearly dependent ആണ്.

Linearly independent: $c_1 = \dots = c_n = 0$ ആകുമ്പോൾ മാത്രം $c_1x_1 + \dots + c_nx_n = 0$ ശരിയാകുന്നുവെങ്കിൽ അവ linearly independent ആണ്.

Vectors $x_1, \dots, x_n$ are **linearly dependent** if there exist scalars $c_1, \dots, c_n$, not all zero, with $c_1x_1 + \dots + c_nx_n = 0$. If the only solution is $c_1 = \dots = c_n = 0$, they are **linearly independent**. "Linear" means only addition and scalar multiplication are used. If vectors are dependent, any vector with a non-zero coefficient can be written as a linear combination of the others.

മലയാളത്തിൽ

Dependent എന്നാൽ കൂട്ടത്തിലെ ഒരു vector മറ്റുള്ളവയിൽ നിന്ന് ഉണ്ടാക്കാം, അതായത് അത് "അധികം" ആണ്. ഒരു recipe-യിൽ "ചായ", "പാൽ", "പാൽച്ചായ" എന്ന് മൂന്നും വെച്ചാൽ മൂന്നാമത്തേത് ആദ്യ രണ്ടിൽ നിന്ന് ഉണ്ടാക്കാം, പുതിയ വിവരം ഇല്ല. Independent എന്നാൽ ഒന്നിനെയും മറ്റുള്ളവയിൽ നിന്ന് ഉണ്ടാക്കാൻ പറ്റില്ല. Test ചെയ്യാൻ: $c_1x_1 + \dots = 0$ എന്ന് എഴുതി equations solve ചെയ്യുക; എല്ലാ c -യും 0 മാത്രമാണ് ഉത്തരം എങ്കിൽ independent.

Dependent: $c_1x_1 + c_2x_2 + \dots + c_nx_n = 0$ with some $c_i \neq 0$

Independent: $c_1x_1 + \dots + c_nx_n = 0 \Rightarrow c_1 = c_2 = \dots = c_n = 0$

(a) If a subset $x_1, \dots, x_k$ ($k < n$) is dependent, the whole set $x_1, \dots, x_n$ is dependent.

(b) If $x_1, \dots, x_n$ are independent, every subset is independent.

Any set containing the zero vector is dependent.

EXAMPLE 1 (FROM THE UNIT)

Show $a = (1, 2, -1)$, $b = (3, 2, 5)$, $c = (-2, 0, -6)$ are linearly dependent.

Suppose $\alpha a + \beta b + \gamma c = 0$. Comparing entries:

$$\alpha + 3\beta - 2\gamma = 0 \dots (1) \quad 2\alpha + 2\beta = 0 \dots (2) \quad -\alpha + 5\beta - 6\gamma = 0 \dots (3)$$

From (2), $\beta = -\alpha$. Put in (1): $\alpha - 3\alpha - 2\gamma = 0 \Rightarrow \gamma = -\alpha$. Check (3): $-\alpha - 5\alpha + 6\alpha = 0 \checkmark$.

Take $\alpha = 1$, $\beta = -1$, $\gamma = -1$ (not all zero): $a - b - c = (1 - 3 + 2, 2 - 2 - 0, -1 - 5 + 6) = (0, 0, 0)$. Hence dependent (and $a = b + c$).

EXAMPLE 2 (FROM THE UNIT)

(i) $x = (1, 0, 3)$, $y = (3, 0, 9)$. (ii) $x = (1, 3)$, $y = (-2, 5)$, $z = (3, -2)$.

(i) $3x - y = (3 - 3, 0, 9 - 9) = 0$, so dependent; $y = 3x$, a scalar multiple of x .

(ii) $x - y = (1 + 2, 3 - 5) = (3, -2) = z$, so $x - y - z = 0$ and the vectors are dependent; z is a linear combination of x and y .

EXAMPLE 3

Are $(1, 2)$ and $(3, 4)$ linearly independent?

$$c_1(1, 2) + c_2(3, 4) = (0, 0) \text{ gives } c_1 + 3c_2 = 0 \text{ and } 2c_1 + 4c_2 = 0.$$

From the first, $c_1 = -3c_2$; then $-6c_2 + 4c_2 = 0 \Rightarrow c_2 = 0 \Rightarrow c_1 = 0$. Only the zero solution, so they are **independent**. (Geometrically, two vectors in $\mathbb{R}^2$ are dependent only if they lie on one line through the origin.)

EXAMPLE 4 (FROM THE UNIT)

Show that any set containing the zero vector is dependent.

Let $a_1 = 0, a_2, \dots, a_n$. Then $1 \cdot a_1 + 0 \cdot a_2 + \dots + 0 \cdot a_n = 0$, and the coefficient $1 \neq 0$. Hence dependent.

Exam tip: the unit's definition in 8.6 says "linearly independent if there exist scalars not all zero"; that first word should be **dependent**. Also, in Example 2(ii) the unit says z is a "scalar multiple" of x and y ; it is a linear combination ($z = x - y$).

Generators (spanning sets)

Definition: A set of vectors generates (spans) a space if every vector of the space can be written as a linear combination of vectors in the set.

Space-ലെ ഓരോ vector-നെയും ഒരു set-ലെ vectors-ന്റെ linear combination ആയി എഴുതാൻ കഴിയുമെങ്കിൽ, ആ set ആ space-നെ generate (span) ചെയ്യുന്നു.

In the Euclidean space E_n (all real n -tuples with addition, scalar multiplication and distance), a set of vectors **generates** (spans) the space if every vector can be written as a linear combination of them. In E_2 , $u = (1, 0)$ and $v = (0, 1)$ generate everything: $(x_1, x_2) = x_1u + x_2v$. Key points: an arbitrary choice of vectors may not generate the space; E_n needs at least n generators; and generating sets are not unique.

മലയാളത്തിൽ

Generators എന്നാൽ "അടിസ്ഥാന ചേരുവകൾ": ഇവ ശരിയായ അളവിൽ ചേർത്ത് space-ലെ ഏത് vector-ഉം ഉണ്ടാക്കാം. Paint mix ചെയ്ത് ഏത് നിറവും ഉണ്ടാക്കുന്നത് പോലെ. $\mathbb{R}^2$ -ൽ ഒരൊറ്റ vector കൊണ്ട് ഒരു line മാത്രമേ കിട്ടൂ, മുഴുവൻ plane കിട്ടാൻ കുറഞ്ഞത് രണ്ട് (ഒരേ line-ൽ അല്ലാത്ത) vectors വേണം.

$\{a_1, \dots, a_k\}$ generates (spans) $E_n \Leftrightarrow$ every $x \in E_n$ is $x = c_1a_1 + \dots + c_ka_k$ for some $c_i \in \mathbb{R}$

$E_2: (x_1, x_2) = x_1(1, 0) + x_2(0, 1)$

E_n needs at least n generators

EXAMPLE 1 (FROM THE UNIT)

Express $(2, 5)$ using different generating sets.

With $(1, 0), (0, 1)$: $2(1, 0) + 5(0, 1) = (2, 5)$.

With $(1, 1), (0, 1)$: $2(1, 1) + 3(0, 1) = (2, 2) + (0, 3) = (2, 5)$.

With $(1, 0), (1, 1), (0, 1)$: $(1, 0) + (1, 1) + 4(0, 1) = (2, 5)$.

With $(1, 2), (0, 1)$: $2(1, 2) + 1(0, 1) = (2, 4) + (0, 1) = (2, 5)$.

So the same vector can be generated in many ways, and generating sets are not unique.

EXAMPLE 2

Does the single vector $(1, 2)$ generate $\mathbb{R}^2$? Do $(1, 2)$ and $(2, 4)$ generate $\mathbb{R}^2$?

$c(1, 2) = (c, 2c)$ always has second entry twice the first, so $(1, 0)$ cannot be made. No.

$(2, 4) = 2(1, 2)$, so every combination is still of the form $(t, 2t)$: only the line $y = 2x$. No; two vectors on the same line do not generate the plane.

Exam tip: the extracted unit text lists a generating set as $(1, 2), (1, 1), (0, 0)$ with the combination $2(1, 2) + 1(0, 1) + 3(0, 1)$, which gives $(2, 8)$, not $(2, 5)$. The consistent version is $2(1, 2) + 1(0, 1) = (2, 5)$, shown above.

Basis

Definition: A basis of a vector space is a set of vectors that is linearly independent and generates (spans) the space.

ഒരു vector space-ന്റെ basis എന്നാൽ linearly independent ആയതും ആ space-നെ generate (span) ചെയ്യുന്നതുമായ vectors-ന്റെ set.

A **basis** of a space is a set of vectors that (i) generates the space and (ii) is linearly independent. Any two linearly independent vectors of E_2 form a basis of E_2 : they are enough to generate everything, and any extra vector would be unnecessary. In a basis every vector is written in exactly one way (uniquely). The standard basis of E_3 is $e_1 = (1, 0, 0), e_2 = (0, 1, 0), e_3 = (0, 0, 1)$.

മലയാളത്തിൽ

Basis എന്നാൽ ഏറ്റവും കുറഞ്ഞ, അധികം ഇല്ലാത്ത generators set. എല്ലാം ഉണ്ടാക്കാൻ പറ്റണം (span), പക്ഷേ ഒന്നും വെറുതെ ആവരുത് (independent). ഒരു കൂട്ടുകാരന് വഴി പറയുമ്പോൾ "കിഴക്കോട്ട് 2 km, വടക്കോട്ട് 5 km" എന്ന് രണ്ട് ദിശ മതി, മൂന്നാമതൊരു ദിശ ആവശ്യമില്ല. Subspace-ന്റെ basis കണ്ടുപിടിക്കാൻ: condition ഉപയോഗിച്ച് ഒരു variable ഒഴിവാക്കി, general vector-നെ free variables-ന്റെ combination ആയി split ചെയ്യുക.

Basis of V = linearly independent set that generates V

Standard basis of E_3 : $e_1 = (1, 0, 0), e_2 = (0, 1, 0), e_3 = (0, 0, 1)$

$(a_1, a_2, a_3) = a_1e_1 + a_2e_2 + a_3e_3$ (unique)

Method for a subspace: solve the condition $\rightarrow$ write general vector $\rightarrow$ split by free variables $\rightarrow$ check independence

EXAMPLE 1 (FROM THE UNIT)

Write $a = (2, 3, 4)$ in terms of the standard basis of E_3 .

$a = 2(1, 0, 0) + 3(0, 1, 0) + 4(0, 0, 1) = 2e_1 + 3e_2 + 4e_3$, and this is the only way.

EXAMPLE 2

Show $\{(1, 1), (1, -1)\}$ is a basis of $\mathbb{R}^2$ and write $(2, 5)$ in it.

Independent: $c_1(1, 1) + c_2(1, -1) = 0 \Rightarrow c_1 + c_2 = 0, c_1 - c_2 = 0 \Rightarrow c_1 = c_2 = 0 \checkmark$.

Generates: for (x, y) solve $a + b = x, a - b = y \Rightarrow a = (x + y)/2, b = (x - y)/2$, always possible $\checkmark$.

For $(2, 5)$: $a = 7/2, b = -3/2$. Check: $(7/2)(1, 1) - (3/2)(1, -1) = (7/2 - 3/2, 7/2 + 3/2) = (2, 5) \checkmark$.

EXAMPLE 3 (CHECK YOUR PROGRESS)

Find a basis for $W = \{(x, y, z) \mid 2x + y - z = 0\}$.

Step 1: $z = 2x + y$, so $(x, y, z) = (x, y, 2x + y)$.

Step 2: split: $(x, 0, 2x) + (0, y, y) = x(1, 0, 2) + y(0, 1, 1)$. So $(1, 0, 2), (0, 1, 1)$ generate W .

Step 3: independence: $x(1, 0, 2) + y(0, 1, 1) = (x, y, 2x + y) = (0, 0, 0) \Rightarrow x = 0, y = 0 \checkmark$.

Basis: $\{(1, 0, 2), (0, 1, 1)\}$.

EXAMPLE 4 (CHECK YOUR PROGRESS)

Find a basis for $W = \{(x, y, z) \mid x = 2y = 3z\}$.

Let $x = 2y = 3z = k$. Then $x = k, y = k/2, z = k/3$, so $(x, y, z) = (k/6)(6, 3, 2)$. Equivalently $x/6 = y/3 = z/2$.

Every vector of W is a multiple of $(6, 3, 2)$, and a single non-zero vector is independent. Basis: $\{(6, 3, 2)\}$.

Check: $6 = 2 \cdot 3 = 3 \cdot 2 \checkmark$.

Exam tip: the unit's answer key prints the basis for $x = 2y = 3z$ as $\{(b, 3, 2)\}$; it should be $\{(6, 3, 2)\}$, as the hint $x/6 = y/3 = z/2$ shows. For a basis question always show both parts: "generates" and "independent".

Quick recap

- A vector is an ordered n -tuple of real numbers; order matters. The null vector has all entries 0.
- Add/subtract entry-wise (same length only); λa multiplies every entry by λ .
- Geometrically a vector is an arrow from the origin to the point (a_1, a_2, a_3) ; addition follows the parallelogram law.
- Inner product $\langle a, b \rangle = \sum a_i b_i$ is a number, not a vector.
- Norm $\|x\| = \sqrt{\langle x, x \rangle} = \sqrt{(\sum x_i^2)}$; remember the square root.
- Orthogonal $\Leftrightarrow \langle a, b \rangle = 0$; $\cos \theta = \langle a, b \rangle / (\|a\| \|b\|)$; Schwarz's inequality keeps $|\cos \theta| \leq 1$.
- A vector space is closed under addition and scalar multiplication and obeys the 9 axioms (commutative, associative, zero, negative, distributive laws, $1u = u$).
- Subspace test: $0 \in W$, closed under addition, closed under scalar multiplication.

- In $\mathbb{R}^2$ the subspaces include the axes and every line through the origin; lines not through the origin are not subspaces.
- Dependent: some non-trivial combination gives 0; independent: only the trivial one does.
- Any set containing 0 is dependent; a superset of a dependent set is dependent; a subset of an independent set is independent.
- Generators (spanning set): every vector is a linear combination of them; not unique; E_n needs at least n .
- Basis = independent + generating; representation in a basis is unique. Standard basis of E_3 : e_1, e_2, e_3 .
- The unit's summary also mentions eigenvalues and matrices, but those are not covered in this unit's text.

Check yourself

Q. 1. If $x_1 = (2, 9, 8)$, $x_2 = (0, 1, 0)$, $x_3 = (1, 0, 1)$, find $2x_2 + 5x_3 - x_1$.

$$2x_2 = (0, 2, 0); 5x_3 = (5, 0, 5).$$

$$(0 + 5 - 2, 2 + 0 - 9, 0 + 5 - 8) = (3, -7, -3).$$

Q. 2. Find the norm of (i) $(2, 5)$, (ii) $(-2, 2)$, (iii) $(1, -2, 2)$.

$$(i) \sqrt{4 + 25} = \sqrt{29}. (ii) \sqrt{4 + 4} = \sqrt{8} = 2\sqrt{2}. (iii) \sqrt{1 + 4 + 4} = \sqrt{9} = 3.$$

(The unit's key says 9 and 8 for (i) and (ii); those are wrong: $\sqrt{29}$ and $2\sqrt{2}$ are correct.)

Q. 3. Find $\langle a, b \rangle$ for (i) $(2, 3, 4)$, $(4, 5, 5)$ and (ii) $(-2, -3, 4)$, $(4, 5, -6)$.

$$(i) 8 + 15 + 20 = 43. (ii) -8 - 15 - 24 = -47.$$

Q. 4. Find the angle between $a = (1, 1, 0)$ and $b = (1, 0, 1)$.

$$\langle a, b \rangle = 1 + 0 + 0 = 1; \|a\| = \sqrt{2}, \|b\| = \sqrt{2}.$$

$$\cos \theta = 1/(\sqrt{2} \cdot \sqrt{2}) = 1/2 \Rightarrow \theta = \pi/3 (60^\circ).$$

Q. 5. For what value of k are $(2, k, 1)$ and $(3, -1, 4)$ orthogonal?

$$\langle a, b \rangle = 6 - k + 4 = 10 - k = 0 \Rightarrow k = 10. \text{ Check: } 6 - 10 + 4 = 0 \checkmark.$$

Q. 6. Is $W = \{(x, y) \mid x + y = 1\}$ a subspace of $\mathbb{R}^2$? Is $W = \{(x, y) \mid x + y = 0\}$?

First: $(0, 0)$ gives $0 + 0 = 0 \neq 1$, so $0 \notin W$: **not** a subspace.

Second: $0 \in W$; $(a, -a) + (b, -b) = (a + b, -(a + b)) \in W$; $\alpha(a, -a) = (\alpha a, -\alpha a) \in W$. It **is** a subspace (the line $y = -x$).

Q. 7. Test $(1, 2, 3)$ and $(2, 4, 6)$ for linear dependence. Then test $(1, 0, 1)$, $(0, 1, 1)$, $(1, 1, 0)$.

First pair: $2(1, 2, 3) - (2, 4, 6) = 0$ with coefficients $2, -1$ not zero: **dependent**.

Second set: $c_1(1, 0, 1) + c_2(0, 1, 1) + c_3(1, 1, 0) = 0$ gives $c_1 + c_3 = 0$, $c_2 + c_3 = 0$, $c_1 + c_2 = 0$. So $c_1 = c_2 = -c_3$, and then $-2c_3 = 0 \Rightarrow$ all zero: **independent**.

Q. 8. Express $(4, -1)$ as a combination of $(1, 2)$ and $(3, 1)$. Do these two vectors form a basis of $\mathbb{R}^2$?

$a(1, 2) + b(3, 1) = (4, -1)$: $a + 3b = 4$, $2a + b = -1$. From the first, $a = 4 - 3b$; then $8 - 6b + b = -1 \Rightarrow b = 9/5$, $a = 4 - 27/5 = -7/5$.

Check: $(-7/5 + 27/5, -14/5 + 9/5) = (4, -1) \checkmark$.

They are independent (neither is a multiple of the other) and two independent vectors in $\mathbb{R}^2$ form a basis: **yes**.

Q. 9. Find a basis for $W = \{(x, y, z) \mid x = 2y = 3z\}$.

$x/6 = y/3 = z/2 = t \Rightarrow (x, y, z) = t(6, 3, 2)$. Basis: $\{(6, 3, 2)\}$ (the key's "(b, 3, 2)" is a misprint).

Q. 10. Find a basis for $W = \{(x, y, z) \mid x + y + z = 0\}$.

$z = -x - y \Rightarrow (x, y, z) = x(1, 0, -1) + y(0, 1, -1)$, so these two generate W .

Independence: $x(1, 0, -1) + y(0, 1, -1) = (x, y, -x - y) = 0 \Rightarrow x = y = 0 \checkmark$.

Basis: $\{(1, 0, -1), (0, 1, -1)\}$.

UNIT 9

Linear Spaces-II

ഈ unit-ൽ ഒരു linear equations system-നെ $AX = d$ എന്ന matrix equation ആയി എഴുതുന്നതും, eigen value, eigen vector, diagonalisation എന്നിവ കണ്ടുപിടിക്കുന്നതും പഠിക്കുന്നു. പിന്നെ quadratic forms, അവയുടെ definiteness, vectors-നെ വെച്ചുള്ള differentiation (linear, quadratic, bilinear forms) എന്നിവയും നോക്കുന്നു.

Vectors and matrices: $AX = d$

The characteristic value problem $Ax = \lambda x$

Characteristic equation

Sum and product of roots, properties

Characteristic vectors (eigen vectors)

Diagonalisation

Linear independence of eigen vectors

Quadratic forms

Definiteness and eigen values

Vector differentiation of a linear function

Vector differentiation of a vector of functions

Vector differentiation of a quadratic form

Vector differentiation of a bilinear form

Vectors and matrices: $AX = d$

Definition: A system of m linear equations in n unknowns is written as the matrix equation $AX = d$, where $A = [a_{ij}]$ is the $m \times n$ coefficient matrix, X the $n \times 1$ vector of unknowns and d the $m \times 1$ vector of constants.

n unknowns ഉള്ള m linear equations-ന്റെ system-നെ $AX = d$ എന്ന matrix equation ആയി എഴുതുന്നു; ഇവിടെ $A = [a_{ij}]$ $m \times n$ coefficient matrix, X unknowns-ന്റെ $n \times 1$ vector, d constants-ന്റെ $m \times 1$ vector.

In this unit a vector is an n -tuple $(x_1, x_2, \dots, x_n)$ of real numbers; the vector with all elements zero is the **null vector**. A system of m linear equations in n unknowns has three parts: the coefficients a_{ij} , the variables x_j and the constants d_i . Put each part into its own array and the whole system becomes one matrix equation $AX = d$, where A is $m \times n$, X is $n \times 1$ and d is $m \times 1$.

മലയാളത്തിൽ

ഒരുപാട് equations ഒന്നിച്ച് എഴുതുന്നതിന് പകരം, coefficients എല്ലാം ഒരു box-ൽ (A), variables വേറൊരു column-ൽ (X), constants മറ്റൊരു column-ൽ (d) എന്ന് അടുക്കി വെക്കുന്നു. കടയിലെ bill പോലെ: ഓരോ സാധനത്തിന്റെ വിലയും (A) എണ്ണവും (X) ഗുണിച്ചാൽ total (d) കിട്ടും. A-യുടെ i-ാം row-ഉം X-ഉം ഗുണിച്ചാൽ i-ാം equation-ന്റെ ഇടത്തെ വശം തന്നെ കിട്ടും.

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = d_1$$

⋮

$$a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = d_m$$

$$\Leftrightarrow AX = d, \text{ with } A = [a_{ij}]_{m \times n}, X = [x_j]_{n \times 1}, d = [d_i]_{m \times 1}$$

EXAMPLE (FROM THE UNIT)

Write $6x_1 + 3x_2 + x_3 = 22$, $x_1 + 4x_2 - 2x_3 = 12$, $4x_1 - x_2 + 5x_3 = 10$ as a matrix equation.

Row i of A holds the coefficients of equation i (a missing variable gets 0; " x_3 " means coefficient 1):

$$\begin{pmatrix} 6 & 3 & 1 \\ 1 & 4 & -2 \\ 4 & -1 & 5 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 22 \\ 12 \\ 10 \end{pmatrix}$$

Check: row 1 $\times$ X = $6x_1 + 3x_2 + 1 \cdot x_3$, which is the left side of equation 1.

EXAMPLE

Write $2x - y = 3$, $x + 5y = 7$ in the form $AX = d$.

$$A = \begin{pmatrix} 2 & -1 \\ 1 & 5 \end{pmatrix}, X = \begin{pmatrix} x \\ y \end{pmatrix}, d = \begin{pmatrix} 3 \\ 7 \end{pmatrix}. \text{ Here } m = n = 2, \text{ so } A \text{ is } 2 \times 2.$$

The characteristic value problem $Ax = \lambda x$

Eigen value: a scalar λ is an eigen (characteristic) value of a square matrix A if $Ax = \lambda x$ for some non-null vector x .

Eigen vector: such a non-null x is an eigen (characteristic) vector of A belonging to λ .

Eigen value: ഏതെങ്കിലും ഒരു non-null vector x -ന് $Ax = \lambda x$ ആണെങ്കിൽ, scalar λ square matrix A -യുടെ eigen (characteristic) value ആണ്.

Eigen vector: അങ്ങനെയുള്ള non-null x , λ -ക്ക് അനുബന്ധമായ A -യുടെ eigen (characteristic) vector ആണ്.

Given an $n \times n$ matrix A , we look for a scalar λ and a **non-null** vector x such that $Ax = \lambda x$. Then A acting on x gives back a scalar multiple of x : the direction of x is unchanged, only its length is stretched (or flipped) by λ . λ is called the **characteristic value** (characteristic root, eigen value, proper value) and x the **characteristic vector** (eigen vector) of A .

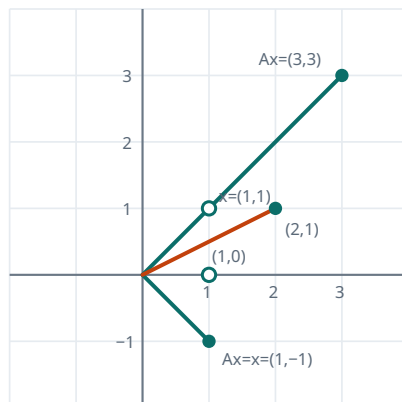
മലയാളത്തിൽ

സാധാരണ ഒരു matrix ഒരു vector-നെ തിരിക്കുകയും നീട്ടുകയും ചെയ്യും. പക്ഷേ ചില പ്രത്യേക vectors ഉണ്ട്, അവയെ A ഒട്ടും തിരികില്ല, വെറുതെ λ മടങ്ങ് നീട്ടുകയോ ചുരുക്കുകയോ മാത്രം ചെയ്യും. അവയാണ് eigen vectors, ആ λ ആണ് eigen value. കാറ്റിക്ലിനോൾ തെങ്ങിന്റെ ഓലകൾ മിക്കതും ദിശ മാറും, പക്ഷേ കാറ്റിന്റെ അതേ ദിശയിൽ നിൽക്കുന്ന ഓല ദിശ മാറാതെ അങ്ങനെ തന്നെ നിൽക്കും; eigen vector-ഉം അതുപോലെയാണ്. $x = 0$ എപ്പോഴും $Ax = \lambda x$ satisfy ചെയ്യും, അതുകൊണ്ട് അത് eigen vector ആയി കണക്കാക്കില്ല.

$$Ax = \lambda x, x \neq 0$$

λ : eigen value (characteristic root)

x : eigen vector (characteristic vector) belonging to λ



$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$. Along $(1, 1)$ A stretches by 3 ($\lambda = 3$); along $(1, -1)$ it changes nothing ($\lambda = 1$). The vector $(1, 0)$ is turned to a new direction $(2, 1)$, so it is not an eigen vector.

EXAMPLE

Let $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$. Check whether $(1, 1)$, $(1, -1)$ and $(1, 0)$ are eigen vectors.

$A(1, 1)$: $(2 \cdot 1 + 1 \cdot 1, 1 \cdot 1 + 2 \cdot 1) = (3, 3) = 3(1, 1)$. Eigen vector, $\lambda = 3$.

$A(1, -1)$: $(2 - 1, 1 - 2) = (1, -1) = 1(1, -1)$. Eigen vector, $\lambda = 1$.

$A(1, 0)$: $(2, 1)$. Is $(2, 1) = \lambda(1, 0)$? That needs $2 = \lambda$ and $1 = 0$, impossible. Not an eigen vector.

Characteristic equation

Characteristic polynomial: $|A - \lambda I|$, a polynomial of degree n in λ , for an $n \times n$ matrix A .

Characteristic equation: $|A - \lambda I| = 0$; its roots are the eigen values of A .

Characteristic polynomial: $n \times n$ matrix A -ക്ക്, λ -ൽ degree n ഉള്ള polynomial ആയ $|A - \lambda I|$.

Characteristic equation: $|A - \lambda I| = 0$; ഇതിന്റെ roots ആണ് A -യുടെ eigen values.

Rewrite $Ax = \lambda x$ as $(A - \lambda I)x = 0$, a homogeneous system. It has a non-trivial solution $x \neq 0$ only when $\text{rank}(A - \lambda I) < n$, that is, when $|A - \lambda I| = 0$. Expanding this determinant gives a polynomial of degree n in λ , the **characteristic polynomial**; setting it to zero gives the **characteristic equation**. Its n roots (real or complex, distinct or repeated, zero or non-zero) are the eigen values.

മലയാളത്തിൽ

$Ax = \lambda x$ എന്നത് $(A - \lambda I)x = 0$ എന്ന് മാറ്റി എഴുതാം. ഈ system-ന് $x = 0$ അല്ലാത്ത ഒരു solution വേണമെങ്കിൽ $A - \lambda I$ singular ആകണം, അതായത് അതിന്റെ determinant പൂജ്യം ആകണം. അതുകൊണ്ട് ആദ്യം $|A - \lambda I| = 0$ solve ചെയ്ത് λ കണ്ടുപിടിക്കുന്നു, പിന്നെ മാത്രമേ x നോക്കൂ. $n \times n$ matrix ആണെങ്കിൽ λ -യിൽ degree n ഉള്ള equation കിട്ടും, അതിന് n roots ഉണ്ടാകും.

$$(A - \lambda I)x = 0 \text{ has } x \neq 0 \Leftrightarrow |A - \lambda I| = 0$$

$$\begin{aligned}\text{Order 2: } |A - \lambda I| &= (a_{11} - \lambda)(a_{22} - \lambda) - a_{12}a_{21} \\ &= \lambda^2 - (a_{11} + a_{22})\lambda + (a_{11}a_{22} - a_{12}a_{21}) = 0\end{aligned}$$

EXAMPLE (FROM THE UNIT)

Find the eigen values of $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$.

$$|A - \lambda I| = \begin{vmatrix} 2 - \lambda & 1 \\ 1 & 2 - \lambda \end{vmatrix} = (2 - \lambda)^2 - 1.$$

$$\text{Set to 0: } 4 - 4\lambda + \lambda^2 - 1 = 0 \Rightarrow \lambda^2 - 4\lambda + 3 = 0 \Rightarrow (\lambda - 3)(\lambda - 1) = 0.$$

$$\text{So } \lambda_1 = 3, \lambda_2 = 1.$$

EXAMPLE (FROM THE UNIT)

Find all eigen values of $A = \begin{pmatrix} 2 & 1 & 3 \\ 1 & -1 & 0 \\ 1 & 2 & 3 \end{pmatrix}$.

$$|A - \lambda I| = \begin{vmatrix} 2 - \lambda & 1 & 3 \\ 1 & -1 - \lambda & 0 \\ 1 & 2 & 3 - \lambda \end{vmatrix}. \text{ Expand along } R_1:$$

$$= (2 - \lambda)[(-1 - \lambda)(3 - \lambda) - 0 \cdot 2] - 1[1 \cdot (3 - \lambda) - 0 \cdot 1] + 3[1 \cdot 2 - (-1 - \lambda) \cdot 1]$$

$$(-1 - \lambda)(3 - \lambda) = \lambda^2 - 2\lambda - 3, \text{ so the first term is } (2 - \lambda)(\lambda^2 - 2\lambda - 3) = -\lambda^3 + 4\lambda^2 - \lambda - 6.$$

$$\text{Second term: } -(3 - \lambda) = -3 + \lambda. \text{ Third term: } 3(3 + \lambda) = 9 + 3\lambda.$$

$$\text{Add: } -\lambda^3 + 4\lambda^2 + 3\lambda + 0 = 0 \Rightarrow \lambda^3 - 4\lambda^2 - 3\lambda = 0 \Rightarrow \lambda(\lambda^2 - 4\lambda - 3) = 0.$$

$$\lambda = 0 \text{ or } \lambda = (4 \pm \sqrt{(16 + 12)})/2 = (4 \pm \sqrt{28})/2 = 2 \pm \sqrt{7}.$$

EXAMPLE

Find the eigen values of $A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$.

Use the order-2 formula: $\lambda^2 - (4 + 3)\lambda + (4 \cdot 3 - 1 \cdot 2) = \lambda^2 - 7\lambda + 10 = (\lambda - 5)(\lambda - 2) = 0$.

$\lambda_1 = 5, \lambda_2 = 2$.

Exam tip: subtract λ only from the diagonal entries. Expand along the row or column with the most zeros to save work, and keep the sign pattern $+$ $-$ $+$ for the cofactors.

Sum and product of roots, properties

Trace: the trace of a square matrix is the sum of its principal diagonal elements.

Sum and product of eigen values: $\lambda_1 + \dots + \lambda_n = \text{Trace}(A)$ and $\lambda_1 \lambda_2 \dots \lambda_n = |A|$.

Trace: ഒരു square matrix-ന്റെ principal diagonal elements-ന്റെ തുകയാണ് trace.

Sum and product of eigen values: $\lambda_1 + \dots + \lambda_n = \text{Trace}(A)$, $\lambda_1 \lambda_2 \dots \lambda_n = |A|$.

Write the characteristic equation as $\lambda^n + c_1 \lambda^{n-1} + \dots + c_n = 0$. Then $c_1 = -(\text{sum of diagonal entries})$, $c_2 = \text{sum of all principal minors of order 2}$, and in general $c_j = (-1)^j \times (\text{sum of all principal minors of order } j)$, with $c_n = (-1)^n |A|$. From the relation between roots and coefficients: the sum of the eigen values equals the **trace** of A (sum of the principal diagonal), and their product equals $|A|$.

മലയാളത്തിൽ

Eigen values കണ്ടുപിടിച്ച ശേഷം ഉത്തരം ശരിയാണോ എന്ന് ഉടനെ check ചെയ്യാൻ ഇത് ഉപകരിക്കും: എല്ലാ eigen values-ന്റെയും തുക = diagonal elements-ന്റെ തുക (trace), ഗുണനഫലം = determinant. Principal minor എന്നാൽ ഒരേ നമ്പറുള്ള rows-ഉം columns-ഉം എടുത്ത് ഉണ്ടാക്കുന്ന minor. ഇതുവഴി determinant വലുതായി expand ചെയ്യാതെ തന്നെ 3×3 characteristic equation നേരിട്ട് എഴുതാം.

$$\lambda_1 + \lambda_2 + \dots + \lambda_n = a_{11} + a_{22} + \dots + a_{nn} = \text{Trace}(A)$$

$$\lambda_1 \lambda_2 \dots \lambda_n = |A|$$

$$3 \times 3: \lambda^3 - (\text{Trace } A)\lambda^2 + (\text{sum of order-2 principal minors})\lambda - |A| = 0$$

Property	Statement
(i)	A singular ($ A = 0$) $\Rightarrow$ at least one eigen value is 0 (the product is 0).
(ii)	A real symmetric $\Rightarrow$ all eigen values are real.
(iii)	A non-singular with eigen values $\lambda_i \Rightarrow A^{-1}$ has eigen values $1/\lambda_i$.
(iv)	Diagonal or triangular $A \Rightarrow$ eigen values are the diagonal entries.
(v)	Eigen values of A^k (k a positive integer) are λ_i^k .

EXAMPLE

Check the sum and product rules for $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ (eigen values 3, 1).

Trace = $2 + 2 = 4 = 3 + 1$. $|A| = 4 - 1 = 3 = 3 \times 1$. Both agree.

EXAMPLE

Write the characteristic equation of $A = \begin{pmatrix} 1 & 0 & -1 \\ 1 & -1 & -1 \\ -3 & 6 & 3 \end{pmatrix}$ using principal minors.

Trace = $1 + (-1) + 3 = 3$.

Order-2 principal minors: rows/cols {1,2}: $1(-1) - 0 \cdot 1 = -1$; {1,3}: $1 \cdot 3 - (-1)(-3) = 0$; {2,3}: $(-1)(3) - (-1)(6) = 3$. Sum = 2.

$|A| = 1[(-1)(3) - (-1)(6)] - 0 + (-1)[1 \cdot 6 - (-1)(-3)] = 3 - 3 = 0$.

Equation: $\lambda^3 - 3\lambda^2 + 2\lambda - 0 = 0 \Rightarrow \lambda(\lambda - 1)(\lambda - 2) = 0 \Rightarrow \lambda = 0, 1, 2$. Since $|A| = 0$, property (i) predicted a zero root.

EXAMPLE

A 2×2 matrix A has eigen values 2 and 5. Find the eigen values of A^{-1} and A^3 , and $|A|$.

A^{-1} : $1/2$ and $1/5$. A^3 : 8 and 125. $|A| = 2 \times 5 = 10$.

Characteristic vectors (eigen vectors)

Definition: An eigen vector of A for eigen value λ_i is any non-zero solution x of the homogeneous system $(A - \lambda_i I)x = 0$.

Eigen value λ_i -ക്കുള്ള A -യുടെ eigen vector എന്നാൽ $(A - \lambda_i I)x = 0$ എന്ന homogeneous system-ന്റെ പൂജ്യമല്ലാത്ത ഏതൊരു solution x -ഉം.

For each eigen value λ_i , solve the homogeneous system $(A - \lambda_i I)x = 0$. Because $A - \lambda_i I$ is singular, the equations are dependent and there are infinitely many solutions; any non-zero one is an eigen vector. To make it unique we fix one component (say set it to 1) or **normalise** it so that its length is 1.

മലയാളത്തിൽ

ഓരോ λ -ക്കും $(A - \lambda I)x = 0$ എന്ന system solve ചെയ്യണം. ഇവിടെ equations തമ്മിൽ ആശ്രയിച്ചിരിക്കുന്നതുകൊണ്ട് ഒരു free variable വരും, അതിന് ഇഷ്ടമുള്ള ഒരു value (സാധാരണ 1) കൊടുത്താൽ മതി. $(1, 1)$ eigen vector ആണെങ്കിൽ $(2, 2)$, $(5, 5)$ ഒക്കെയും eigen vectors ആണ്, ഒരേ ദിശ തന്നെ. Length 1 ആക്കാൻ vector-നെ അതിന്റെ length കൊണ്ട് ഹരിച്ചാൽ മതി, അതാണ് normalise ചെയ്യൽ.

For each λ_i : solve $(A - \lambda_i I)x = 0$, take $x \neq 0$

Normalised: $x / \|x\|$, where $\|(a, b)\| = \sqrt{a^2 + b^2}$

Check: $Ax = \lambda_i x$

EXAMPLE (FROM THE UNIT, $\mathbb{R}^2$)

Find eigen vectors of $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, $\lambda_1 = 3$, $\lambda_2 = 1$.

$\lambda_1 = 3$: $A - 3I = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}$. With $x = (x_1, x_2)$: $-x_1 + x_2 = 0$ and $x_1 - x_2 = 0$. Both say $x_1 = x_2$ (one real equation, two unknowns). Put $x_1 = 1$: eigen vector $(1, 1)$.

$\lambda_2 = 1$: $A - I = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$. $x_1 + x_2 = 0 \Rightarrow x_1 = -x_2$. Put $x_1 = 1$: eigen vector $(1, -1)$.

Normalised (length $\sqrt{2}$ each): $(1/\sqrt{2}, 1/\sqrt{2})$ and $(1/\sqrt{2}, -1/\sqrt{2})$. Their dot product is $1/2 - 1/2 = 0$, so they are orthogonal.

EXAMPLE (FROM THE UNIT, $\mathbb{R}^3$)

Find eigen vectors of $A = \begin{pmatrix} 1 & 0 & -1 \\ 1 & -1 & -1 \\ -3 & 6 & 3 \end{pmatrix}$ with $\lambda = 0, 1, 2$.

$\lambda = 0$: $Ax = 0$ gives $x - z = 0$, $x - y - z = 0$, $-3x + 6y + 3z = 0$. First: $x = z$. Put into second: $z - y - z = 0 \Rightarrow y = 0$. Third: $-3z + 0 + 3z = 0 \checkmark$. Take $z = 1$: eigen vector $(1, 0, 1)$.

$\lambda = 1$: $(A - I)x = 0$ with $A - I = \begin{pmatrix} 0 & 0 & -1 \\ 1 & -2 & -1 \\ -3 & 6 & 2 \end{pmatrix}$: $-z = 0$, $x - 2y - z = 0$, $-3x + 6y + 2z = 0$. So $z = 0$, then $x = 2y$;

third: $-6y + 6y = 0 \checkmark$. Take $y = 1$: eigen vector $(2, 1, 0)$.

$\lambda = 2$: $A - 2I = \begin{pmatrix} -1 & 0 & -1 \\ 1 & -3 & -1 \\ -3 & 6 & 1 \end{pmatrix}$: $-x - z = 0$, $x - 3y - z = 0$, $-3x + 6y + z = 0$. First: $x = -z$. Second: $-z - 3y - z = 0$

$\Rightarrow 3y = -2z$. Take $z = -3$: $x = 3$, $y = 2$. Third: $-9 + 12 - 3 = 0 \checkmark$. Eigen vector $(3, 2, -3)$.

Check $\lambda = 2$: $A(3, 2, -3) = (3 + 0 + 3, 3 - 2 + 3, -9 + 12 - 9) = (6, 4, -6) = 2(3, 2, -3) \checkmark$.

EXAMPLE

Eigen vectors of $A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$ ($\lambda = 5, 2$).

$\lambda = 5$: $A - 5I = \begin{pmatrix} -1 & 1 \\ 2 & -2 \end{pmatrix} \Rightarrow -x + y = 0 \Rightarrow y = x$. Eigen vector $(1, 1)$.

$\lambda = 2$: $A - 2I = \begin{pmatrix} 2 & 1 \\ 2 & 1 \end{pmatrix} \Rightarrow 2x + y = 0 \Rightarrow y = -2x$. Eigen vector $(1, -2)$. Check: $A(1, -2) = (4 - 2, 2 - 6) = (2, -4) = 2(1, -2) \checkmark$.

Exam tip: in the unit's $\lambda = 2$ working the relation is printed as " $3y = 2z$ "; the correct relation is $3y = -2z$ (from $x = -z$ in $x - 3y - z = 0$). The final vector $(3, 2, -3)$ printed there is correct. The unit also prints the characteristic equation of this matrix as $\lambda(\lambda^3 - 3\lambda + 2) = 0$; it should be $\lambda(\lambda^2 - 3\lambda + 2) = \lambda(\lambda - 1)(\lambda - 2) = 0$, which gives the roots 0, 1, 2 it states. Also, the unit's list for $\lambda = 0$ shows " $x + y - z = 0$ "; the second row of A gives $x - y - z = 0$. Always finish by checking $Ax = \lambda x$.

Diagonalisation

Definition: A square matrix A is diagonalisable if there is an invertible matrix P (whose columns are eigen vectors of A) such that $P^{-1}AP = \Lambda$, a diagonal matrix of the eigen values.

$P^{-1}AP = \Lambda$ (eigen values-ന്റെ diagonal matrix) ആകുന്ന ഒരു invertible matrix P (columns A -യുടെ eigen vectors) ഉണ്ടെങ്കിൽ square matrix A diagonalisable ആണ്.

Put the n eigen vectors as the columns of a matrix P . If these columns are linearly independent, P is invertible and $P^{-1}AP = \Lambda$, the diagonal matrix with the eigen values on its diagonal (in the same order as the columns of P). For a real symmetric matrix the eigen values are real and the eigen vectors are mutually orthogonal, so with normalised eigen vectors P is **orthogonal** ($P^T P = I$) and $P^{-1} = P^T$, giving $P^T A P = \Lambda$.

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Diagonalise ചെയ്യുക എന്നാൽ A -യെ eigen vectors-ന്റെ "കണ്ണിലൂടെ" നോക്കുക. ആ ദിശകളിൽ A വെറുതെ λ കൊണ്ട് ഗുണിക്കുക മാത്രമേ ചെയ്യൂ, അതുകൊണ്ട് ആ basis-ൽ A ഒരു diagonal matrix ആയി മാറുന്നു. Eigen vectors-നെ columns ആക്കി P ഉണ്ടാക്കുക, $P^{-1}AP$ കണ്ടാൽ diagonal-ൽ eigen values വരും. Symmetric matrix ആണെങ്കിൽ P^{-1} കണ്ടുപിടിക്കേണ്ട, P^T തന്നെ മതി.

$P = [x^1 \ x^2 \ \dots \ x^n]$ (eigen vectors as columns)

$P^{-1}AP = \Lambda = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$

A symmetric, columns normalised: $P^T P = I$ and $P^T A P = \Lambda$

EXAMPLE (FROM THE UNIT, SYMMETRIC)

Diagonalise $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$.

$P = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{pmatrix}$. $P^T = P$ here, and $P^T P = (1/2 + 1/2, 1/2 - 1/2; 1/2 - 1/2, 1/2 + 1/2) = I$. So P is orthogonal.

AP : columns are $A(1/\sqrt{2}, 1/\sqrt{2}) = (3/\sqrt{2}, 3/\sqrt{2})$ and $A(1/\sqrt{2}, -1/\sqrt{2}) = (1/\sqrt{2}, -1/\sqrt{2})$, so $AP = \begin{pmatrix} 3/\sqrt{2} & 1/\sqrt{2} \\ 3/\sqrt{2} & -1/\sqrt{2} \end{pmatrix}$.

$P^T(AP) = \begin{pmatrix} 3/2 + 3/2 & 1/2 - 1/2 \\ 3/2 - 3/2 & 1/2 + 1/2 \end{pmatrix} = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix} = \Lambda$.

EXAMPLE (FROM THE UNIT, 3×3)

$$\text{Diagonalise } A = \begin{pmatrix} 1 & 0 & -1 \\ 1 & -1 & -1 \\ -3 & 6 & 3 \end{pmatrix}.$$

$$\text{Eigen vectors } (1, 0, 1), (2, 1, 0), (3, 2, -3) \text{ for } \lambda = 0, 1, 2. \text{ So } P = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 1 & 0 & -3 \end{pmatrix}.$$

$$|P| = 1(-3 - 0) - 2(0 - 2) + 3(0 - 1) = -3 + 4 - 3 = -2 \neq 0, \text{ so } P^{-1} \text{ exists: } P^{-1} = (1/2) \begin{pmatrix} 3 & -6 & -1 \\ -2 & 6 & 2 \\ 1 & -2 & -1 \end{pmatrix}.$$

$$P^{-1}A = (1/2) \begin{pmatrix} 0 & 0 & 0 \\ -2 & 6 & 2 \\ 2 & -4 & -2 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ -1 & 3 & 1 \\ 1 & -2 & -1 \end{pmatrix}.$$

$(P^{-1}A)P$: row 2 $\times$ columns of P : $(-1 + 0 + 1, -2 + 3 + 0, -3 + 6 - 3) = (0, 1, 0)$; row 3: $(1 + 0 - 1, 2 - 2 + 0, 3 - 4 + 3) = (0, 0, 2)$.

$$P^{-1}AP = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix} = \text{diag}(0, 1, 2).$$

EXAMPLE

$$\text{Diagonalise } A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix} \text{ (eigen vectors } (1, 1), (1, -2)).$$

$$P = \begin{pmatrix} 1 & 1 \\ 1 & -2 \end{pmatrix}, |P| = -2 - 1 = -3, P^{-1} = (1/3) \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}.$$

$$AP = \begin{pmatrix} 5 & 2 \\ 5 & -4 \end{pmatrix}, P^{-1}AP = (1/3) \begin{pmatrix} 15 & 0 \\ 0 & 6 \end{pmatrix} = \begin{pmatrix} 5 & 0 \\ 0 & 2 \end{pmatrix}.$$

Exam tip: the unit writes " $\lambda_1 = 0, \lambda_2 = 1, \lambda_3 = 1$ " for the 3×3 example; the correct values are 0, 1, 2, as its own final matrix $\text{diag}(0, 1, 2)$ shows. The order of eigen values in Λ must match the order of eigen vectors in P .

Linear independence of eigen vectors

Definition: Vectors $v_1, \dots, v_k$ are linearly independent if $c_1 v_1 + \dots + c_k v_k = 0$ implies every $c_i = 0$; eigen vectors belonging to distinct eigen values are always linearly independent.

$c_1 v_1 + \dots + c_k v_k = 0$ ആണെങ്കിൽ എല്ലാ $c_i = 0$ ആകണം എങ്കിൽ $v_1, \dots, v_k$ linearly independent ആണ്; distinct eigen values-ന്റെ eigen vectors എപ്പോഴും linearly independent ആണ്.

Vectors $v_1, \dots, v_k$ are **linearly independent** if $c_1 v_1 + \dots + c_k v_k = 0$ forces every $c_i = 0$. If the eigen values $\lambda_1, \dots, \lambda_n$ are distinct, the corresponding eigen vectors are linearly independent, so P is invertible and A can be diagonalised. Distinct roots are sufficient, not necessary: a matrix with a repeated root may still have n independent eigen vectors. For a **symmetric** matrix, eigen vectors of distinct roots are also orthogonal.

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Linearly independent എന്നാൽ ഒരു vector-നെയും ബാക്കിയുള്ളവയുടെ combination ആയി എഴുതാൻ പറ്റില്ല, ഓരോന്നും പുതിയൊരു ദിശ തരുന്നു. Eigen values എല്ലാം വ്യത്യസ്തമാണെങ്കിൽ eigen vectors ഉറപ്പായും independent ആയിരിക്കും, അപ്പോൾ P -ക്ക് inverse ഉണ്ട്. പരിശോധിക്കാൻ $c_1 v_1 + c_2 v_2 + \dots = 0$ എന്ന system solve ചെയ്ത് എല്ലാ c -യും പൂജ്യം മാത്രം ആണോ എന്ന് നോക്കുക, അല്ലെങ്കിൽ $|P| \neq 0$ ആണോ എന്ന് നോക്കുക.

$$c_1 v_1 + c_2 v_2 + \dots + c_n v_n = 0 \Rightarrow c_1 = \dots = c_n = 0 \text{ (independent)}$$

n vectors in $\mathbb{R}^n$ are independent $\Leftrightarrow |P| \neq 0$ (P = the vectors as columns)

Distinct eigen values $\Rightarrow$ independent eigen vectors $\Rightarrow A$ diagonalisable

EXAMPLE ($\mathbb{R}^3$)

Show that the eigen vectors $(1, 0, 1), (2, 1, 0), (3, 2, -3)$ are linearly independent.

Suppose $c_1(1, 0, 1) + c_2(2, 1, 0) + c_3(3, 2, -3) = (0, 0, 0)$. Compare components:

$$x: c_1 + 2c_2 + 3c_3 = 0$$

$$y: c_2 + 2c_3 = 0$$

$$z: c_1 - 3c_3 = 0$$

From z : $c_1 = 3c_3$. From y : $c_2 = -2c_3$. Put into x : $3c_3 - 4c_3 + 3c_3 = 2c_3 = 0 \Rightarrow c_3 = 0$, so $c_1 = 0, c_2 = 0$.

Only the zero solution: independent. (Same as $|P| = -2 \neq 0$.) This is expected, since $\lambda = 0, 1, 2$ are distinct.

EXAMPLE ($\mathbb{R}^2$, REPEATED ROOT)

Compare $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. Both have $\lambda = 1$ repeated.

For I : $(I - I)x = 0$ is $0 = 0$, so every vector is an eigen vector. $(1, 0)$ and $(0, 1)$ are two independent eigen vectors: I is (already) diagonal.

For B : $B - I = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ gives $y = 0$ only, so every eigen vector is a multiple of $(1, 0)$. There is only one independent eigen vector, so B cannot be diagonalised.

EXAMPLE (ORTHOGONALITY NEEDS SYMMETRY)

$A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$ has distinct roots 5, 2 with eigen vectors $(1, 1)$, $(1, -2)$. They are independent ($|P| = -3 \neq 0$), but $(1, 1) \cdot (1, -2) = 1 - 2 = -1 \neq 0$, so they are not orthogonal. A is not symmetric.

Exam tip: the unit says "if two roots are distinct then the characteristic vectors corresponding to them will be orthogonal". This holds only for symmetric matrices. Its own 3×3 example is a counterexample: $(1, 0, 1) \cdot (2, 1, 0) = 2 \neq 0$.

Quadratic forms

Definition: A quadratic form in $x_1, \dots, x_n$ is a homogeneous polynomial of degree 2, $Q(x) = \sum_i \sum_j a_{ij} x_i x_j = x^T A x$, with A symmetric.

$x_1, \dots, x_n$ -ലെ quadratic form എന്നാൽ degree 2 ഉള്ള homogeneous polynomial, $Q(x) = \sum_i \sum_j a_{ij} x_i x_j = x^T A x$, ഇവിടെ A symmetric.

An expression like $a_{11}x_1^2 + 2a_{12}x_1x_2 + a_{22}x_2^2$, where every term has total degree 2, is a **quadratic form**. In n variables it is $Q(x) = \sum_i \sum_j a_{ij} x_i x_j = x^T A x$ with A symmetric. To build A : put the coefficient of x_i^2 at (i, i) , and split the coefficient of $x_i x_j$ equally, half at (i, j) and half at (j, i) . Q is **positive definite** if $Q(x) > 0$ for all $x \neq 0$, **positive semi-definite** if $Q(x) \geq 0$ for all x , similarly negative (semi-)definite, and **indefinite** if it takes both signs.

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Quadratic form എന്നാൽ എല്ലാ terms-ഉം degree 2 ആയ expression: x^2 , xy , y^2 പോലുള്ളവ മാത്രം, constant-ഓ x -ഓ ഒറ്റയ്ക്ക് ഇല്ല. Symmetric matrix ഉണ്ടാക്കുമ്പോൾ xy -യുടെ coefficient രണ്ട് പേർക്ക് തുല്യമായി പങ്കിടുന്നത് പോലെ, പകുതി (1,2)-ലും പകുതി (2,1)-ലും ഇടുക. x , y എന്ത് കൊടുത്താലും Q എപ്പോഴും positive ആണെങ്കിൽ positive definite, എപ്പോഴും negative ആണെങ്കിൽ negative definite, ചിലപ്പോൾ + ചിലപ്പോൾ - ആണെങ്കിൽ indefinite.

$$Q(x_1, x_2) = a_{11}x_1^2 + 2a_{12}x_1x_2 + a_{22}x_2^2 = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{pmatrix} a_{11} & a_{12} \\ a_{12} & a_{22} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

Completing the square: $Q = a_{11}[(x_1 + (a_{12}/a_{11})x_2)^2 + ((a_{11}a_{22} - a_{12}^2)/a_{11}^2)x_2^2]$

Positive definite $\Leftrightarrow a_{11} > 0$ and $a_{11}a_{22} - a_{12}^2 > 0$

Negative definite $\Leftrightarrow a_{11} < 0$ and $a_{11}a_{22} - a_{12}^2 > 0$

EXAMPLE (FROM THE UNIT)

$x_1^2 - 4x_1x_2 + 4x_2^2 = (x_1 - 2x_2)^2 \geq 0$ always, and it is 0 at $(2, 1) \neq (0, 0)$. So it is positive semi-definite (not definite).

$-10x_1^2 + 6x_1x_2 - x_2^2 = -x_1^2 - (3x_1 - x_2)^2$. Check: $(3x_1 - x_2)^2 = 9x_1^2 - 6x_1x_2 + x_2^2$, and $-x_1^2 - 9x_1^2 + 6x_1x_2 - x_2^2$ matches. It is ≤ 0 , and = 0 only when $x_1 = 0$ and $3x_1 = x_2$, i.e. at $(0, 0)$: negative definite.

EXAMPLE (FROM THE UNIT)

Write $Q(x, y) = x^2 + 6xy + 10y^2$ in symmetric matrix form and show $Q(x, y) > 0$ for all $(x, y) \neq (0, 0)$.

Split $6xy = 3xy + 3xy$: $Q = x^2 + 3xy + 3xy + 10y^2 = x(x + 3y) + y(3x + 10y)$.

$$= \begin{bmatrix} x & 3y & 3x & 10y \end{bmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{bmatrix} x & y \end{bmatrix} \begin{pmatrix} 1 & 3 \\ 3 & 10 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

Complete the square: $Q = x^2 + 6xy + 9y^2 + y^2 = (x + 3y)^2 + y^2 \geq 0$.

$Q = 0 \Leftrightarrow x + 3y = 0$ and $y = 0 \Leftrightarrow x = 0, y = 0$. So $Q > 0$ for every $(x, y) \neq (0, 0)$. (Also $a_{11} = 1 > 0$ and $1 \cdot 10 - 3^2 = 1 > 0$.)

EXAMPLE ($\mathbb{R}^3$)

Write $Q = x^2 + 2y^2 - 7z^2 + 4xy + 8yz - 6zx$ in symmetric matrix form.

Diagonal: 1, 2, -7. Halve the cross terms: $xy \rightarrow 2$, $yz \rightarrow 4$, $zx \rightarrow -3$.

$$Q = [x \ y \ z] \begin{pmatrix} 1 & 2 & -3 \\ 2 & 2 & 4 \\ -3 & 4 & -7 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}.$$

Check: row form $(x + 2y - 3z)x + (2x + 2y + 4z)y + (-3x + 4y - 7z)z$ expands back to Q .

Exam tip: the unit's answer key for Check Your Progress 1, Q3 prints the (3, 2) entry of the 3×3 matrix as 2; it must be 4 (half of $8yz$), as above. Also, the unit's worked example for $x^2 + 6xy + 10y^2$ writes " $x^2 + 3xy + 3xy + y^2$ " and the matrix $\begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix}$. That matrix gives $x^2 + 6xy + y^2$, a different (indefinite) form. The correct matrix is $\begin{bmatrix} 1 & 3 \\ 3 & 10 \end{bmatrix}$. Always multiply back to check.

Definiteness and eigen values

Positive (negative) definite: $Q = x^T A x$ is positive (negative) definite if $Q(x) > 0$ (< 0) for all $x \neq 0$, i.e. all eigen values of A are > 0 (< 0).

Positive (negative) semi-definite: $Q(x) \geq 0$ (≤ 0) for all x , i.e. all eigen values are ≥ 0 (≤ 0).

Indefinite: Q takes both positive and negative values, i.e. A has eigen values of both signs.

Positive (negative) definite: എല്ലാ $x \neq 0$ -നും $Q(x) > 0$ (< 0) ആണെങ്കിൽ, അതായത് A -യുടെ എല്ലാ eigen values-ഉം > 0 (< 0) ആണെങ്കിൽ, $Q = x^T A x$ positive (negative) definite ആണ്.

Positive (negative) semi-definite: എല്ലാ x -നും $Q(x) \geq 0$ (≤ 0), അതായത് എല്ലാ eigen values-ഉം ≥ 0 (≤ 0).

Indefinite: Q positive-ഉം negative-ഉം values എടുക്കുന്നു, അതായത് A -ക്ക് രണ്ട് signs-ലുമുള്ള eigen values ഉണ്ട്.

Because a symmetric A can be diagonalised by an orthogonal P , substituting $x = Py$ turns Q into $\lambda_1 y_1^2 + \dots + \lambda_n y_n^2$. So the signs of the eigen values decide the sign of Q . All $\lambda > 0$: positive definite; all $\lambda \geq 0$: positive semi-definite; all $\lambda < 0$: negative definite; all $\lambda \leq 0$: negative semi-definite; mixed signs: indefinite. For 2×2 , the same can be read from a_{11} , a_{22} and $|A|$.

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Eigen vectors-ന്റെ ദിശയിൽ നോക്കിയാൽ Q എന്നത് $\lambda_1 y_1^2 + \lambda_2 y_2^2$ എന്ന ലളിതമായ രൂപത്തിലാകും, squares എപ്പോഴും ≥ 0 ആയതുകൊണ്ട് λ -കളുടെ sign ആണ് Q -ന്റെ sign തീരുമാനിക്കുന്നത്. എല്ലാ eigen values-ഉം positive ആണെങ്കിൽ positive definite, എല്ലാം ≥ 0 ആയിരിക്കുകയും ഒന്നെങ്കിലും 0 ആകുകയും ചെയ്താൽ positive semi-definite, ഒന്ന് + ഉം മറ്റൊന്ന് - ഉം ആണെങ്കിൽ indefinite. 2×2 -ന് eigen values കണ്ടുപിടിക്കാതെ തന്നെ $|A|$, a_{11} നോക്കി പറയാം.

Type	Eigen values	2×2 test ($A = [a_{11} \ a_{12}; a_{12} \ a_{22}]$)
Positive definite	all $\lambda > 0$	$a_{11} > 0, A > 0$
Positive semi-definite	all $\lambda \geq 0$	$a_{11} \geq 0, a_{22} \geq 0, A \geq 0$
Negative definite	all $\lambda < 0$	$a_{11} < 0, A > 0$
Negative semi-definite	all $\lambda \leq 0$	$a_{11} \leq 0, a_{22} \leq 0, A \geq 0$
Indefinite	some $\lambda > 0$, some $\lambda < 0$	$ A < 0$

EXAMPLE

Classify $x^2 - 4xy + 4y^2$ using eigen values.

$$A = \begin{pmatrix} 1 & -2 \\ -2 & 4 \end{pmatrix}. \text{Trace} = 5, |A| = 4 - 4 = 0.$$

$$\lambda^2 - 5\lambda + 0 = 0 \Rightarrow \lambda = 0, 5. \text{ Both } \geq 0, \text{ one is } 0: \text{ positive semi-definite (matches } (x - 2y)^2).$$

EXAMPLE

Classify $x^2 + 6xy + y^2$.

$$A = \begin{pmatrix} 1 & 3 \\ 3 & 1 \end{pmatrix}, |A| = 1 - 9 = -8 < 0 \Rightarrow \text{indefinite.}$$

$$\text{Eigen values: } \lambda^2 - 2\lambda - 8 = (\lambda - 4)(\lambda + 2) = 0 \Rightarrow \lambda = 4, -2 \text{ (mixed signs). Direct check: } Q(1, 0) = 1 > 0, Q(1, -1) = 1 - 6 + 1 = -4 < 0.$$

EXAMPLE

Classify $-10x^2 + 6xy - y^2$.

$$A = \begin{pmatrix} -10 & 3 \\ 3 & -1 \end{pmatrix} : a_{11} = -10 < 0, |A| = 10 - 9 = 1 > 0 \Rightarrow \text{negative definite.}$$

Eigen values: $\lambda^2 + 11\lambda + 1 = 0$. Sum $-11 < 0$ and product $1 > 0$, so both roots are negative. Consistent.

Exam tip: the unit states "Q is positive semi-definite $\Leftrightarrow |A| = 0$ and $a_{11} \geq 0$ ". That describes the semi-definite-but-not-definite case and is incomplete: $A = [0 \ 0; 0 \ -1]$ has $|A| = 0$ and $a_{11} = 0$, yet $Q = -x_2^2 \leq 0$. The full test is $a_{11} \geq 0, a_{22} \geq 0$ and $|A| \geq 0$ (equivalently, both eigen values ≥ 0).

Vector differentiation of a linear function

Definition: For constant a and variable x , the linear function $a'x = a_1x_1 + \dots + a_nx_n$ has vector derivative $\partial(a'x)/\partial x = a$, the vector of its partial derivatives.

Constant a -ക്കും variable x -നും, linear function $a'x = a_1x_1 + \dots + a_nx_n$ -ന്റെ vector derivative $\partial(a'x)/\partial x = a$ ആണ്, അതായത് അതിന്റെ partial derivatives-ന്റെ vector.

For $a = (a_1, \dots, a_n)^T$ and $x = (x_1, \dots, x_n)^T$, the expression $a'x = a_1x_1 + \dots + a_nx_n$ is a **linear function**. Its partial derivative with respect to x_i is simply a_i . Arranging all n partial derivatives as a vector gives $\partial(a'x)/\partial x = a$. This is the vector version of $d(ax)/dx = a$.

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Vector differentiation എന്നാൽ ഓരോ variable-നെ വെച്ചും partial derivative എടുത്ത് അവയെ ഒരു vector ആയി അടുക്കി വെക്കുക. $a'x$ -ൽ x_1 -നെ വെച്ച് differentiate ചെയ്താൽ a_1 മാത്രം ബാക്കി, മറ്റ് terms constant ആയി പോകും. അതുകൊണ്ട് ഉത്തരം a തന്നെ, സാധാരണ $d(5x)/dx = 5$ പോലെ.

$$a'x = a_1x_1 + a_2x_2 + \dots + a_nx_n$$

$$\partial(a'x)/\partial x_i = a_i$$

$$\partial(a'x)/\partial x = a$$

EXAMPLE

$a = (2, -1, 3)^T$, $x = (x_1, x_2, x_3)^T$. Find $\partial(a'x)/\partial x$.

$$a'x = 2x_1 - x_2 + 3x_3. \text{ Partial: } 2, -1, 3. \text{ So } \partial(a'x)/\partial x = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} = a.$$

EXAMPLE (CHECK YOUR PROGRESS 1, Q2)

$a = (2a, -a, 3a, a)^T$, $x = (x_1, x_2, x_3, x_4)^T$. Prove $\partial(a'x)/\partial x = a$.

$a'x = 2ax_1 - ax_2 + 3ax_3 + ax_4$. $\partial/\partial x_1 = 2a$, $\partial/\partial x_2 = -a$, $\partial/\partial x_3 = 3a$, $\partial/\partial x_4 = a$. Stacked: $(2a, -a, 3a, a)^T = a$.

Vector differentiation of a vector of functions

Definition: For $y = (y_1, \dots, y_n)$ with each y_j a function of $x = (x_1, \dots, x_m)$, $\partial y/\partial x$ is the $m \times n$ matrix whose (i, j) entry is $\partial y_j/\partial x_i$.

ഓരോ y_j -ഉം $x = (x_1, \dots, x_m)$ -ന്റെ function ആയ $y = (y_1, \dots, y_n)$ -ന്, $\partial y/\partial x$ എന്നത് (i, j) entry $\partial y_j/\partial x_i$ ആയ $m \times n$ matrix ആണ്.

Let $y = (y_1, \dots, y_n)$ where each $y_j = f_j(x_1, \dots, x_m)$. Then $\partial y/\partial x$ is the $m \times n$ matrix whose (i, j) entry is $\partial y_j/\partial x_i$. In IGNOU's layout, each **row** corresponds to one variable x_i and each **column** to one function y_j .

മലയാളത്തിൽ

ഇവിടെ ഒന്നിലധികം functions ഉണ്ട്. ഓരോ function-നെയും ഓരോ variable-നെ വെച്ചും differentiate ചെയ്ത ഒരു table (matrix) ഉണ്ടാക്കുന്നു. IGNOU layout-ൽ row = ഏത് x വെച്ചാണ് differentiate ചെയ്തത്, column = ഏത് y ആണ്. ഒരു class-ലെ marks register പോലെ: rows-ൽ subjects, columns-ൽ students.

$$(\partial y/\partial x)_{ij} = \partial y_j/\partial x_i, \text{ size } m \times n$$

$$\text{Row } i: (\partial y_1/\partial x_i, \partial y_2/\partial x_i, \dots, \partial y_n/\partial x_i)$$

EXAMPLE

$y_1 = x_1^2 + x_2$, $y_2 = x_1 x_2$. Find $\partial y / \partial x$.

Row x_1 : $\partial y_1 / \partial x_1 = 2x_1$, $\partial y_2 / \partial x_1 = x_2$. Row x_2 : $\partial y_1 / \partial x_2 = 1$, $\partial y_2 / \partial x_2 = x_1$.

$$\partial y / \partial x = \begin{pmatrix} 2x_1 & x_2 \\ 1 & x_1 \end{pmatrix}.$$

EXAMPLE (FROM THE UNIT)

$y_1 = x_1^2 + 2x_2^2 - 3x_1 x_3$, $y_2 = 3x_1 x_2^2 - x_2^2 + 4x_2 x_3^2$. Find $\partial y / \partial x$ for $x = (x_1, x_2, x_3)$.

Row x_1 : $\partial y_1 / \partial x_1 = 2x_1 - 3x_3$; $\partial y_2 / \partial x_1 = 3x_2^2$.

Row x_2 : $\partial y_1 / \partial x_2 = 4x_2$; $\partial y_2 / \partial x_2 = 6x_1 x_2 - 2x_2 + 4x_3^2$.

Row x_3 : $\partial y_1 / \partial x_3 = -3x_1$; $\partial y_2 / \partial x_3 = 8x_2 x_3$.

$$\partial y / \partial x = \begin{pmatrix} 2x_1 - 3x_3 & 3x_2^2 \\ 4x_2 & 6x_1 x_2 - 2x_2 + 4x_3^2 \\ -3x_1 & 8x_2 x_3 \end{pmatrix} (3 \times 2).$$

Vector differentiation of a quadratic form

Definition: For a symmetric matrix A , the derivative of the quadratic form $x^T A x$ with respect to x is $\partial(x^T A x) / \partial x = 2Ax$ (as a column; $2x^T A$ as a row).

Symmetric matrix A -ക്ക്, quadratic form $x^T A x$ -ന്റെ x -നെ അപേക്ഷിച്ചുള്ള derivative $\partial(x^T A x) / \partial x = 2Ax$ ആണ് (column ആയി; row ആയി $2x^T A$).

For a symmetric $n \times n$ matrix A , $x^T A x = a_{11}x_1^2 + 2a_{12}x_1x_2 + \dots + 2a_{1n}x_1x_n + a_{22}x_2^2 + \dots + a_{nn}x_n^2$. Differentiating with respect to x_i gives $2(a_{i1}x_1 + \dots + a_{in}x_n)$, which is twice the i -th element of Ax . So $\partial(x^T A x) / \partial x = 2Ax$ as a column, or $2x^T A$ as a row. Choose the form whose shape matches the equation you are working with.

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ഇത് $d(ax^2)/dx = 2ax$ എന്നതിന്റെ matrix version ആണ്. x_1 -നെ വെച്ച് differentiate ചെയ്യുമ്പോൾ x_1^2 term-ൽ നിന്നും എല്ലാ $x_1 x_j$ terms-ൽ നിന്നും ഓരോ ഭാഗം കിട്ടും, അവ ചേർത്താൽ Ax -ന്റെ ആദ്യ element-ന്റെ ഇരട്ടി. A symmetric ആയിരിക്കണം എന്ന് ഓർക്കുക. Column vector-ഉം row vector-ഉം തമ്മിൽ equate ചെയ്യാൻ പറ്റില്ല, അതുകൊണ്ട് സന്ദർഭം അനുസരിച്ച് $2Ax$ അല്ലെങ്കിൽ $2x^T A$ ഉപയോഗിക്കുക.

$$\partial(x^T Ax)/\partial x_i = 2(a_{i1}x_1 + a_{i2}x_2 + \dots + a_{in}x_n)$$

$$\partial(x^T Ax)/\partial x = 2Ax \text{ (column)} = (2x^T A)^T, A \text{ symmetric}$$

EXAMPLE ($\mathbb{R}^2$)

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}. \text{ Find } \partial(x^T Ax)/\partial x \text{ directly and check it equals } 2Ax.$$

$$x^T Ax = 2x_1^2 + 2x_1x_2 + 3x_2^2.$$

$$\partial/\partial x_1 = 4x_1 + 2x_2; \partial/\partial x_2 = 2x_1 + 6x_2.$$

$$2Ax = 2 \begin{pmatrix} 2x_1 + x_2 \\ x_1 + 3x_2 \end{pmatrix} = \begin{pmatrix} 4x_1 + 2x_2 \\ 2x_1 + 6x_2 \end{pmatrix}. \text{ Same.}$$

EXAMPLE (CHECK YOUR PROGRESS 2, Q1)

$$A = \begin{pmatrix} 3 & 1 & -2 \\ 1 & 0 & 3 \\ -2 & 3 & 2 \end{pmatrix}, x = (x_1, x_2, x_3)^T. \text{ Show } \partial(x^T Ax)/\partial x = 2Ax.$$

$$x^T Ax = 3x_1^2 + 0x_2^2 + 2x_3^2 + 2(1)x_1x_2 + 2(-2)x_1x_3 + 2(3)x_2x_3 = 3x_1^2 + 2x_3^2 + 2x_1x_2 - 4x_1x_3 + 6x_2x_3.$$

$$\partial/\partial x_1 = 6x_1 + 2x_2 - 4x_3; \partial/\partial x_2 = 2x_1 + 6x_3; \partial/\partial x_3 = -4x_1 + 6x_2 + 4x_3.$$

$$Ax = (3x_1 + x_2 - 2x_3, x_1 + 3x_3, -2x_1 + 3x_2 + 2x_3)^T; \text{ doubling gives exactly the three partials above.}$$

Exam tip: Check Your Progress 2, Q1 asks to show the derivative equals " $2xA^T$ ". The shapes don't work (x is 3×1 , A^T is 3×3); the intended result is $2Ax$ (or $2x^T A$ as a row), which for symmetric A is the same as $2A^T x$. Also the unit's expansion prints the second term as " $a_{12}x_1x_2$ "; it should be $2a_{12}x_1x_2$.

Vector differentiation of a bilinear form

Bilinear form: $x^T Bz = \sum_i \sum_j b_{ij} x_i z_j$, linear in x for fixed z and linear in z for fixed x .

Its derivatives: $\partial(x^T Bz)/\partial x = Bz$ and $\partial(x^T Bz)/\partial z = B^T x$.

Bilinear form: $x^T Bz = \sum_i \sum_j b_{ij} x_i z_j$; z fixed ആണെങ്കിൽ x -ൽ linear, x fixed ആണെങ്കിൽ z -ൽ linear.

Its derivatives: $\partial(x^T Bz)/\partial x = Bz$, $\partial(x^T Bz)/\partial z = B^T x$.

An expression $x^T B z$, where x^T is $1 \times m$, B is $m \times n$ and z is $n \times 1$, is a **bilinear**

form: it is linear in x when z is fixed, and linear in z when x is fixed. Treat it as a linear function of each vector in turn: with z fixed, $x^T B z = x^T (Bz)$, so $\partial/\partial x = Bz$; with x fixed, $x^T B z = (B^T x)^T z$, so $\partial/\partial z = B^T x$.

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Bilinear form-ൽ രണ്ട് vectors ഉണ്ട്, x -ഉം z -ഉം. x -നെ വെച്ച് differentiate ചെയ്യുമ്പോൾ z -നെ constant ആയി കാണുക, അപ്പോൾ ഇത് $a^T x$ പോലൊരു linear function ആണ്, $a = Bz$. z -നെ വെച്ച് ചെയ്യുമ്പോൾ x constant, ഉത്തരം $B^T x$. Transpose എവിടെ വരും എന്ന് ഓർക്കാൻ: ഉത്തരത്തിന്റെ size differentiate ചെയ്യുന്ന vector-ന്റെ size തന്നെ ആയിരിക്കണം.

$$\begin{aligned} x^T B z &= \sum_i \sum_j b_{ij} x_i z_j \\ \partial(x^T B z)/\partial x &= Bz \quad (m \times 1) \\ \partial(x^T B z)/\partial z &= B^T x \quad (n \times 1) \end{aligned}$$

EXAMPLE ($\mathbb{R}^2$)

$$B = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, x = (x_1, x_2)^T, z = (z_1, z_2)^T.$$

$$x^T B z = x_1 z_1 + 2x_1 z_2 + 3x_2 z_1 + 4x_2 z_2.$$

$$\partial/\partial x_1 = z_1 + 2z_2, \partial/\partial x_2 = 3z_1 + 4z_2: \text{ this is } Bz = (z_1 + 2z_2, 3z_1 + 4z_2)^T.$$

$$\partial/\partial z_1 = x_1 + 3x_2, \partial/\partial z_2 = 2x_1 + 4x_2: \text{ this is } B^T x \text{ with } B^T = \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix}.$$

EXAMPLE (CHECK YOUR PROGRESS 2, Q2)

$$x = (x_1, x_2, x_3)^T, B = \begin{pmatrix} 2b_1 & b_1 + b_2 \\ b_2 + 2b_3 & 3b_2 \\ b_1 + b_3 & 4b_3 \end{pmatrix}, z = (z_1, z_2)^T. \text{ Verify } \partial/\partial x = Bz \text{ and } \partial/\partial z = B^T x.$$

$$x^T B z = x_1[2b_1 z_1 + (b_1 + b_2)z_2] + x_2[(b_2 + 2b_3)z_1 + 3b_2 z_2] + x_3[(b_1 + b_3)z_1 + 4b_3 z_2].$$

$$\partial/\partial x_i \text{ is the bracket next to } x_i, \text{ and the brackets are exactly the rows of } B \text{ times } z: \partial/\partial x = Bz.$$

$$\partial/\partial z_1 = 2b_1 x_1 + (b_2 + 2b_3)x_2 + (b_1 + b_3)x_3 \text{ (column 1 of } B \text{ dotted with } x); \partial/\partial z_2 = (b_1 + b_2)x_1 + 3b_2 x_2 + 4b_3 x_3 \text{ (column 2 of } B \text{ dotted with } x). \text{ Together this is } B^T x.$$

Exam tip: a quadratic form is the special case $z = x$, $B = A$ symmetric: adding the two bilinear derivatives $Ax + A^T x = 2Ax$ gives the quadratic-form rule.

Quick recap

- A linear system is written $AX = d$: A ($m \times n$) holds coefficients, X ($n \times 1$) the unknowns, d ($m \times 1$) the constants.
- Eigen value problem: $Ax = \lambda x$ with $x \neq 0$; A only stretches an eigen vector, never turns it.
- Eigen values solve the characteristic equation $|A - \lambda I| = 0$ (degree n in λ).
- 2×2 : $\lambda^2 - (\text{Trace } A)\lambda + |A| = 0$. 3×3 : $\lambda^3 - (\text{Trace } A)\lambda^2 + (\text{sum of order-2 principal minors})\lambda - |A| = 0$.
- Sum of eigen values = trace; product = $|A|$; singular A has a zero eigen value.
- Triangular/diagonal A : eigen values are the diagonal entries; A^{-1} has $1/\lambda$, A^k has λ^k .
- Eigen vectors: solve $(A - \lambda I)x = 0$, set a free variable to 1, check $Ax = \lambda x$.
- P = eigen vectors as columns $\Rightarrow P^{-1}AP = \text{diag}(\lambda_1, \dots, \lambda_n)$; symmetric A : P orthogonal, $P^TAP = \Lambda$.
- Distinct eigen values $\Rightarrow$ independent eigen vectors (sufficient, not necessary). Orthogonal only if A is symmetric.
- Quadratic form $Q = x^T Ax$: squares on the diagonal, half of each cross-term coefficient off the diagonal.
- 2×2 : positive definite $\Leftrightarrow a_{11} > 0$ and $|A| > 0$; negative definite $\Leftrightarrow a_{11} < 0$ and $|A| > 0$; $|A| < 0 \Rightarrow$ indefinite.
- Definiteness from eigen values: all > 0 PD, all ≥ 0 PSD, all < 0 ND, all ≤ 0 NSD, mixed signs indefinite.
- $\partial(a^T x)/\partial x = a$; $\partial(x^T Ax)/\partial x = 2Ax$ (A symmetric); $\partial(x^T Bz)/\partial x = Bz$, $\partial(x^T Bz)/\partial z = B^T x$.
- $\partial y/\partial x$ for a vector of functions: row i = derivatives with respect to x_i , column j = function y_j .

Check yourself

Q. 1. Define eigen value, eigen vector and characteristic equation of a square matrix A .

A scalar λ is an eigen value of A if there is a non-null vector x with $Ax = \lambda x$; such an x is an eigen vector belonging to λ . The equation $|A - \lambda I| = 0$, a polynomial equation of degree n in λ , is the characteristic equation; its roots are the eigen values.

Q. 2. Find the eigen values and eigen vectors of $A = \begin{bmatrix} 3 & 2 \\ 1 & 2 \end{bmatrix}$. Check with trace and determinant.

$$\lambda^2 - 5\lambda + (6 - 2) = \lambda^2 - 5\lambda + 4 = (\lambda - 4)(\lambda - 1) = 0 \Rightarrow \lambda = 4, 1.$$

$$\lambda = 4: A - 4I = \begin{pmatrix} -1 & 2 \\ 1 & -2 \end{pmatrix} \Rightarrow -x + 2y = 0 \Rightarrow x = 2y \Rightarrow (2, 1).$$

$$\lambda = 1: A - I = \begin{pmatrix} 2 & 2 \\ 1 & 1 \end{pmatrix} \Rightarrow x + y = 0 \Rightarrow (1, -1).$$

Check: $4 + 1 = 5 = 3 + 2$ (trace); $4 \times 1 = 4 = 6 - 2$ ($|A|$). $A(2, 1) = (8, 4) = 4(2, 1) \checkmark$.

Q. 3. Without solving, find the sum and product of the eigen values of $A = \begin{bmatrix} 2 & 1 & 3 \\ 1 & -1 & 0 \\ 1 & 2 & 3 \end{bmatrix}$. What does the product tell you?

Sum = trace = $2 - 1 + 3 = 4$. Product = $|A| = 2(-3 - 0) - 1(3 - 0) + 3(2 + 1) = -6 - 3 + 9 = 0$.

Product 0 $\Rightarrow$ A is singular and one eigen value is 0. Indeed the roots are 0 and $2 \pm \sqrt{7}$, and $0 + (2 + \sqrt{7}) + (2 - \sqrt{7}) = 4$.

Q. 4. The eigen values of a 2×2 matrix A are 2 and 3. Find the eigen values of A^{-1} and A^2 , and $|A|$.

A^{-1} : $1/2, 1/3$. A^2 : 4, 9. $|A| = 2 \times 3 = 6$.

Q. 5. Diagonalise $A = \begin{bmatrix} 3 & 2 \\ 1 & 2 \end{bmatrix}$ using the eigen vectors from Q2.

$$P = \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}, |P| = -2 - 1 = -3, P^{-1} = (1/3) \begin{pmatrix} 1 & 1 \\ 1 & -2 \end{pmatrix}.$$

$$AP = \begin{pmatrix} 8 & 1 \\ 4 & -1 \end{pmatrix}; P^{-1}AP = (1/3) \begin{pmatrix} 12 & 0 \\ 0 & 3 \end{pmatrix} = \begin{pmatrix} 4 & 0 \\ 0 & 1 \end{pmatrix}.$$

Q. 6. Are $(1, 2, 0)$, $(0, 1, 1)$, $(1, 3, 1)$ linearly independent?

$c_1(1, 2, 0) + c_2(0, 1, 1) + c_3(1, 3, 1) = 0$ gives $c_1 + c_3 = 0$, $2c_1 + c_2 + 3c_3 = 0$, $c_2 + c_3 = 0$.

$c_1 = -c_3$, $c_2 = -c_3$; second equation: $-2c_3 - c_3 + 3c_3 = 0$, true for every c_3 . Take $c_3 = 1$: $-(1, 2, 0) - (0, 1, 1) + (1, 3, 1) = 0$.

A non-zero solution exists, so they are **dependent**: $(1, 3, 1) = (1, 2, 0) + (0, 1, 1)$. Such vectors could not be the eigen vectors of three distinct eigen values.

Q. 7. Write in symmetric matrix form: (i) $3x^2 + 6xy + 11y^2$, (ii) $5x^2 + 20xy + 8y^2$, (iii) $x^2 + 12xy + 5y^2$. Classify each.

$$(i) \begin{pmatrix} 3 & 3 \\ 3 & 11 \end{pmatrix}: a_{11} = 3 > 0, |A| = 33 - 9 = 24 > 0 \Rightarrow \text{positive definite.}$$

$$(ii) \begin{pmatrix} 5 & 10 \\ 10 & 8 \end{pmatrix}: |A| = 40 - 100 = -60 < 0 \Rightarrow \text{indefinite.}$$

$$(iii) \begin{pmatrix} 1 & 6 \\ 6 & 5 \end{pmatrix}: |A| = 5 - 36 = -31 < 0 \Rightarrow \text{indefinite.}$$

Q. 8. Write in symmetric matrix form: (i) $6x^2 + 3y^2 + 3z^2 - 4xy - 2yz + 4zx$, (ii) $x^2 + 2y^2 - 6xy + 20xz$, (iii) $3x^2 + 3y^2 + 4xy + 6zx + 10yz$.

$$(i) \begin{pmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{pmatrix}$$

$$(ii) \begin{pmatrix} 1 & -3 & 10 \\ -3 & 2 & 0 \\ 10 & 0 & 0 \end{pmatrix} \text{ (no } z^2 \text{ term, no } yz \text{ term).}$$

$$(iii) \begin{pmatrix} 3 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 0 \end{pmatrix}$$

Note: the unit's answer key prints (2, 1) entry of (ii) as 3; it must be -3 so that the matrix is symmetric and the xy coefficient is $-3 - 3 = -6$.

Q. 9. Using eigen values, classify $Q = 2x^2 + 2xy + 2y^2$.

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \text{ eigen values } 3 \text{ and } 1 \text{ (found earlier). Both } > 0 \Rightarrow \text{positive definite.}$$

With the orthogonal P of the diagonalisation example, $Q = 3y_1^2 + y_2^2$, clearly > 0 for $y \neq 0$.

Q. 10. For $A = \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}$ and $x = (x_1, x_2)^T$, find $\partial(x^T Ax)/\partial x$ from the expanded form and verify it equals $2Ax$. Also find $\partial(a'x)/\partial x$ for $a = (4, -7)^T$.

$$x^T Ax = x_1^2 + 4x_1x_2 + 5x_2^2. \partial/\partial x_1 = 2x_1 + 4x_2, \partial/\partial x_2 = 4x_1 + 10x_2.$$

$$2Ax = 2(x_1 + 2x_2, 2x_1 + 5x_2)^T = (2x_1 + 4x_2, 4x_1 + 10x_2)^T \checkmark.$$

$$a'x = 4x_1 - 7x_2 \Rightarrow \partial(a'x)/\partial x = (4, -7)^T = a.$$

UNIT 10

Techniques of Counting and Binomial Theorem

ഓരോ സാധ്യതയും എഴുതി നോക്കാതെ തന്നെ "എത്ര വഴികളുണ്ട്?" എന്ന് എണ്ണാനുള്ള വിദ്യകളാണ് ഈ unit-ൽ: factorial, multiplication/addition principles, permutation (ക്രമം പ്രധാനം), combination (തിരഞ്ഞെടുപ്പ് മാത്രം). അവസാനം combination ഉപയോഗിച്ച് $(a + b)^n$ expand ചെയ്യുന്ന Binomial Theorem-ഉം പഠിക്കുന്നു.

Factorial and its notation

Fundamental Principle of Multiplication (FPM)

Fundamental Principle of Addition (FPA)

Linear permutation: ${}^n P_r$

Permutations with repetition and with alike things

Circular permutation

Combination: ${}^n C_r$

Permutation or combination? Properties of ${}^n C_r$ and problems

Some important results: particular things included or excluded

Binomial theorem for positive integral index

Binomial theorem for rational index

Factorial and its notation

Definition: For a natural number n , n factorial is $n! = n \times (n - 1) \times \dots \times 2 \times 1$, the product of the first n natural numbers; by definition $0! = 1$.

Natural number n -ന്, n factorial എന്നത് $n! = n \times (n - 1) \times \dots \times 2 \times 1$, അതായത് ആദ്യത്തെ n natural numbers-ന്റെ ഗുണനഫലം; definition പ്രകാരം $0! = 1$.

The product of the first n natural numbers is written **$n!$** (read " n factorial"): $n! = n \times (n - 1) \times \dots \times 3 \times 2 \times 1$. By definition **$0! = 1$** . The most useful property is that a factorial can be "opened" as far as you need: $n! = n \times (n - 1)! = n \times (n - 1) \times (n - 2)!$, and so on. This is how large factorials cancel in fractions.

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$n!$ എന്നാൽ 1 മുതൽ n വരെയുള്ള എല്ലാ natural numbers-ന്റെയും ഗുണനഫലം. $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$. വലിയ factorials കണ്ടാൽ പേടിക്കേണ്ട: $10!/8!$ എന്നതിൽ $10!$ -നെ $10 \times 9 \times 8!$ എന്ന് തുറന്നെഴുതിയാൽ $8!$ വെട്ടിപ്പോകും, ബാക്കി 90 മാത്രം. $0! = 1$ എന്നത് ഒരു definition ആണ്, ഓർത്തുവെക്കുക; permutation, combination formulas ശരിയായി പ്രവർത്തിക്കാൻ ഇത് ആവശ്യമാണ്.

$$n! = n(n-1)(n-2) \dots 3 \cdot 2 \cdot 1, \quad 0! = 1$$

$$n! = n \cdot (n-1)! \text{ for } n \geq 1$$

$$n! = n(n-1) \cdot (n-2)! \text{ for } n \geq 2, \text{ and so on}$$

$$\text{Values: } 1! = 1, 2! = 2, 3! = 6, 4! = 24, 5! = 120, 6! = 720, 7! = 5040, 8! = 40320$$

EXAMPLE 1

Evaluate (i) $8!$ (ii) $4! \cdot 3!$ (iii) $10!/(8! \cdot 4!)$ (iv) $5! + 4!$ (v) $6! - 4!$

$$(i) 8! = 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 40320$$

$$(ii) 4! \cdot 3! = 24 \times 6 = 144$$

$$(iii) 10!/(8! \cdot 4!) = (10 \times 9 \times 8!)/(8! \times 24) = 90/24 = 15/4$$

$$(iv) 5! + 4! = 120 + 24 = 144$$

$$(v) 6! - 4! = 720 - 24 = 696$$

Note: $4! \cdot 3! = 144$ but $(4 \times 3)! = 12!$ is huge. Factorials do not multiply like ordinary numbers.

EXAMPLE 2: EXPRESS IN FACTORIALS

(i) $5 \cdot 6 \cdot 7 \cdot 8 \cdot 9 \cdot 10$ (ii) $4 \cdot 8 \cdot 12 \cdot 16 \cdot 20 \cdot 24$ (iii) $1 \cdot 3 \cdot 5 \cdot 7 \cdot 9 \cdot 11$ (iv) $2n \cdot 4n \cdot 6n \cdot 8n$

(i) Multiply and divide by the missing $1 \cdot 2 \cdot 3 \cdot 4$: $= 10!/4!$

(ii) Take 4 out of each of the 6 factors: $= 4^6(1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6) = 4^6 \cdot 6!$

(iii) Put in the missing even numbers: $= 11!/(2 \cdot 4 \cdot 6 \cdot 8 \cdot 10) = 11!/(2^5 \cdot 5!) = 11!/(32 \cdot 5!)$

(iv) Take $2n$ out of each of the 4 factors: $= (2n)^4(1 \cdot 2 \cdot 3 \cdot 4) = (2n)^4 \cdot 4!$

EXAMPLE 3: SOLVE FOR $n \in \mathbb{N}$

(i) $(n+2)! = 42 \cdot n!$ (ii) $1/8! + 1/9! = n/10!$ (iii) $2 \cdot n!/(n-2)! = (n+1)!/(n-1)!$

(i) $(n+2)(n+1)n! = 42n! \Rightarrow (n+2)(n+1) = 42 \Rightarrow n^2 + 3n - 40 = 0 \Rightarrow (n+8)(n-5) = 0$. Since $n \in \mathbb{N}$, $n = 5$.
(Check: $7 \times 6 = 42$.)

(ii) Multiply every term by $10!$: $10!/8! + 10!/9! = n \Rightarrow 90 + 10 = n \Rightarrow n = 100$.

(iii) Open both sides: $2n(n-1)(n-2)!/(n-2)! = (n+1)n(n-1)!/(n-1)! \Rightarrow 2n(n-1) = n(n+1) \Rightarrow n^2 - 3n = 0 \Rightarrow n = 0 \text{ or } 3$. Since $n \in \mathbb{N}$, $n = 3$.

Exam tip: In "solve for n" questions, always open the bigger factorial down to the smaller one, cancel, and get a quadratic. Reject negative roots and 0 (n must be a natural number), and write one line saying why.

Fundamental Principle of Multiplication (FPM)

Definition: If one job can be done in m ways and, after it, a second job can be done in n ways, then the two jobs in succession can be done in $m \times n$ ways.

ഒരു job m വിധത്തിലും, അതിനുശേഷം രണ്ടാമത്തെ job n വിധത്തിലും ചെയ്യാമെങ്കിൽ, രണ്ട് jobs-ഉം ഒന്നിനുപിറകെ ഒന്നായി $m \times n$ വിധത്തിൽ ചെയ്യാം.

A counting problem is an experiment done in several stages, where the number of outcomes at each stage does not depend on what happened earlier. **FPM:** if one job can be done in m ways and, after it, a second job in n ways, then both jobs together (one followed by the other) can be done in $m \times n$ ways. This extends to any number of jobs done in succession. Keyword: **AND** $\Rightarrow$ multiply.

മലയാളത്തിൽ

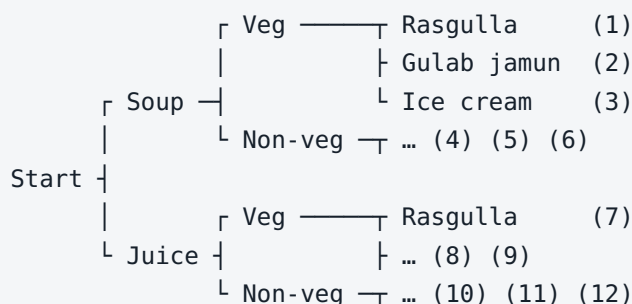
ഒരു ജോലി പല ഘട്ടങ്ങളായി ചെയ്യണം, ഓരോ ഘട്ടവും നിർബന്ധമാണ് എങ്കിൽ ഓരോ ഘട്ടത്തിലെയും options ഗുണിക്കുക. ഒരു ഹോട്ടലിൽ 2 തരം ചായ, 3 തരം പലഹാരം ഉണ്ടെങ്കിൽ "ചായയും പലഹാരവും" എന്ന combo $2 \times 3 = 6$ വിധത്തിൽ ഓർഡർ ചെയ്യാം. സൈക്കിളിലെ 3-wheel number lock-ൽ ഓരോ wheel-നും 0 മുതൽ 9 വരെ 10 options, അപ്പോൾ ആകെ $10 \times 10 \times 10 = 1000$ codes. "ഉം" (AND) വന്നാൽ ഗുണിക്കുക.

Two jobs in succession: $m \times n$ ways

n jobs in succession: $m_1 \times m_2 \times m_3 \times \dots \times m_n$ ways

EXAMPLE 4: GUEST-HOUSE LUNCH (TREE DIAGRAM)

Appetizer: soup or juice (2). Main course: veg or non-veg (2). Dessert: rasgulla, gulab jamun or ice cream (3).
How many complete meals are possible?



Three stages with 2, 2, 3 choices. By FPM, total = $2 \times 2 \times 3 = 12$ meals. Option 7 is juice $\rightarrow$ veg $\rightarrow$ rasgulla; option 12 is juice $\rightarrow$ non-veg $\rightarrow$ ice cream.

EXAMPLE 5

A college has 40 male and 30 female teachers. The principal must pick one male AND one female teacher to go with students on a picnic. In how many ways?

Job 1: pick a male teacher (40 ways). Job 2: pick a female teacher (30 ways). Both jobs must be done, so by FPM: $40 \times 30 = 1200$ ways.

EXAMPLE 7: NUMBER PLATES

How many number plates can be made with two different English letters followed by four different digits?

Fill six places in order: 1st letter 26 ways, 2nd letter 25 ways (one letter used), then digits 10, 9, 8, 7 ways.
Total = $26 \times 25 \times 10 \times 9 \times 8 \times 7 = 32,76,000$ (3276000).

If digits may repeat (as in real plates) but the number 0000 is not issued: $26 \times 25 \times (10^4 - 1) = 26 \times 25 \times 9999$.

EXAMPLE: KERALA REGISTRATION NUMBERS

A new series in Ernakulam uses plates of the form KL-07 AB 1234: "KL-07" is fixed, then two letters (repetition allowed, e.g. AA), then a number from 0001 to 9999. How many plates can one district issue in this format?

"KL-07" gives only 1 way. Letters: $26 \times 26 = 676$. Number: 9999 ways (0000 not issued).
Total = $1 \times 676 \times 9999 = 67,59,324$.

EXAMPLE: A COMMON TRAP

How many 3-digit numbers have all digits different?

The hundreds place cannot be 0, so it has 9 choices (1–9). The tens place can be 0 but not the digit already used: 9 choices. Units place: 8 choices.

Total = $9 \times 9 \times 8 = 648$.

Exam tip: Fill the most restricted position first (the first digit that cannot be 0, the letter that must be a vowel, etc.), then the rest. Always state "by FPM" in the answer.

Fundamental Principle of Addition (FPA)

Definition: If one job can be done in m ways and another in n ways, and only one of the two (they cannot happen together) is to be done, it can be done in $m + n$ ways.

ഒരു job m വിധത്തിലും മറ്റൊന്ന് n വിധത്തിലും ചെയ്യാമെങ്കിൽ, ഒരുമിച്ച് നടക്കാൻ കഴിയാത്ത ഇവയിൽ ഒന്ന് മാത്രം ചെയ്യേണ്ടപ്പോൾ അത് $m + n$ വിധത്തിൽ ചെയ്യാം.

FPA: if one job can be done in m ways and another, independent job in n ways, and we have to do only one of them (they cannot happen together), then there are $m + n$ ways. It extends to any number of mutually exclusive jobs. Keyword: **OR** $\Rightarrow$ add.

മലയാളത്തിൽ

രണ്ടിൽ ഏതെങ്കിലും ഒന്ന് മാത്രം ചെയ്യാൽ മതി എങ്കിൽ options കൂട്ടുക.

തിരുവനന്തപുരത്ത് നിന്ന് കോഴിക്കോട് പോകാൻ 5 train-ഉം 8 KSRTC bus-ഉം ഉണ്ടെങ്കിൽ, "train അല്ലെങ്കിൽ bus" എന്നതിൽ ഒന്ന് തിരഞ്ഞെടുക്കാൻ $5 + 8 = 13$ വഴികൾ. ഒരേസമയം രണ്ടിലും യാത്ര ചെയ്യാൻ പറ്റില്ലല്ലോ; അതാണ് "mutually exclusive". "അല്ലെങ്കിൽ" (OR) വന്നാൽ കൂട്ടുക.

One of two jobs: $m + n$ ways

One of n jobs (no two can happen together): $m_1 + m_2 + \dots + m_n$ ways

Rule of thumb: AND $\Rightarrow$ multiply, OR $\Rightarrow$ add

EXAMPLE 6

The same college (40 male, 30 female teachers) needs one teacher, male or female, for exam duty. In how many ways?

Either pick a male (40 ways) OR a female (30 ways). By FPA: $40 + 30 = 70$ ways.

EXAMPLE: FPM AND FPA TOGETHER

A KSRTC depot has 4 Fast Passenger buses and 3 Super Fast buses to Kottayam. A passenger goes to Kottayam by one of these buses and comes back by a bus of a different type. In how many ways?

Case 1: go by Fast Passenger AND return by Super Fast: $4 \times 3 = 12$.

Case 2: go by Super Fast AND return by Fast Passenger: $3 \times 4 = 12$.

Case 1 OR case 2: $12 + 12 = 24$ ways.

Exam tip: Break a problem into cases joined by OR (add them), and inside each case the steps joined by AND (multiply them). This one habit solves most "at least" and "either/or" problems.

Linear permutation: ${}^n P_r$

Permutation: an arrangement of things in a definite order.

${}^n P_r$: the number of permutations of n different things taken r at a time: ${}^n P_r = n!/(n-r)!$, $0 \leq r \leq n$.

Permutation: ഒരു നിശ്ചിത ക്രമത്തിലുള്ള വസ്തുക്കളുടെ arrangement.

${}^n P_r$: n വ്യത്യസ്ത വസ്തുക്കളിൽ നിന്ന് ഒരേ സമയം r എണ്ണം എടുത്തുള്ള permutations-ന്റെ എണ്ണം: ${}^n P_r = n!/(n-r)!$, $0 \leq r \leq n$.

A **permutation** is an arrangement, so order matters: AB and BA are different. Arrangements in a line are linear permutations. The number of permutations of n different things taken r at a time ($0 \leq r \leq n$) is written ${}^n P_r$ or $P(n, r)$. It comes straight from FPM: the first place can be filled in n ways, the second in $n-1$ ways, ..., the r th in $n-r+1$ ways.

മലയാളത്തിൽ

Permutation എന്നാൽ ക്രമീകരണം (arrangement); ആര് എവിടെ ഇരിക്കുന്നു എന്നത് പ്രധാനം. ഒരു KSRTC bus-ൽ 8 ഒഴിഞ്ഞ സീറ്റുകൾ, 5 യാത്രക്കാർ കയറുന്നു എന്ന് കരുതുക. ആദ്യത്തെയാൾക്ക് 8 സീറ്റ്, രണ്ടാമന് 7, മൂന്നാമന് 6, നാലാമന് 5, അഞ്ചാമന് 4. ആകെ $8 \times 7 \times 6 \times 5 \times 4 = {}^8 P_5 = 6720$ വിധം. "Red, Green, White" പുസ്തകങ്ങൾ RG എന്നും GR എന്നും വെച്ചാൽ രണ്ട് വ്യത്യസ്ത arrangements ആണ്.

$${}^n P_r = n!/(n-r)! = n(n-1)(n-2) \dots (n-r+1) \quad (r \text{ factors}), 0 \leq r \leq n$$

$${}^n P_0 = 1, \quad {}^n P_1 = n, \quad {}^n P_2 = n(n-1), \quad {}^n P_n = n!, \quad {}^n P_{n-1} = n!$$

Three books R, G, W: taken 1 at a time: R, G, W ($3 = {}^3 P_1$); 2 at a time: RG, GR, RW, WR, GW, WG ($6 = {}^3 P_2$); all 3: $6 = {}^3 P_3$

EXAMPLE 8

Evaluate (i) ${}^8 P_2$ (ii) ${}^{20} P_5$ (iii) $P(10, 4)$ (iv) $P(8, 8)$ (v) ${}^5 P_0$

(i) $8 \times 7 = 56$

(ii) $20 \times 19 \times 18 \times 17 \times 16 = 18,60,480$

(iii) $10 \times 9 \times 8 \times 7 = 5040$

(iv) $8!/0! = 40320$

(v) $5!/5! = 1$

EXAMPLE 9

Find n if ${}^n P_5 = 30 \cdot {}^n P_3$.

$$n!/(n-5)! = 30 \cdot n!/(n-3)! \Rightarrow (n-3)!/(n-5)! = 30 \Rightarrow (n-3)(n-4) = 30$$

$$\Rightarrow n^2 - 7n - 18 = 0 \Rightarrow (n-9)(n+2) = 0 \Rightarrow n = 9 \quad (n \text{ cannot be } -2).$$

Check: ${}^9 P_5 = 15120$ and $30 \times {}^9 P_3 = 30 \times 504 = 15120$.

EXAMPLE 10

How many words (with or without meaning) can be formed using all the letters of EQUATION?

8 letters, all different, all used: ${}^8 P_8 = 8! = 40320$.

EXAMPLE 11

How many signals can be made with one or more of 4 flags of different colours (flags hoisted one above another)?

1 flag: ${}^4 P_1 = 4$; 2 flags: ${}^4 P_2 = 12$; 3 flags: ${}^4 P_3 = 24$; 4 flags: ${}^4 P_4 = 24$.

1 OR 2 OR 3 OR 4 flags, so add: $4 + 12 + 24 + 24 = 64$.

EXAMPLE: QUEUE AND RANKING

(a) In how many ways can 5 students stand in a queue? (b) 10 lottery tickets are in a box; first, second and third prizes are drawn one after another. How many results are possible?

(a) ${}^5 P_5 = 5! = 120$.

(b) The prizes are different, so the order of drawing matters: ${}^{10} P_3 = 10 \times 9 \times 8 = 720$.

Exam tip: nP_r needs $r \leq n$. If an equation gives, say, $n = 9$ in 5P_n , reject it: you cannot arrange 9 things out of 5.

Permutations with repetition and with alike things

Repetition allowed: the number of arrangements of r places filled from n different things, each usable any number of times, is n^r .

Alike things: the number of permutations of n things, of which p_1 are alike of one kind, p_2 of another, ..., p_k of the k th, is $n!/(p_1! p_2! \dots p_k!)$.

Repetition allowed: n വ്യത്യസ്ത വസ്തുക്കൾ എത്ര തവണ വേണമെങ്കിലും ഉപയോഗിച്ച് r places നിറയ്ക്കാനുള്ള arrangements-ന്റെ എണ്ണം n^r .

Alike things: n വസ്തുക്കളിൽ p_1 എണ്ണം ഒരു തരത്തിലും, p_2 മറ്റൊരു തരത്തിലും, ..., p_k k th തരത്തിലും ഒരുപോലെയാണെങ്കിൽ, permutations-ന്റെ എണ്ണം $n!/(p_1! p_2! \dots p_k!)$.

Two variations come up constantly. (1) **Repetition allowed:** if each of r places can be filled by any of n things (a thing may be used again), the count is $n \times n \times \dots \times n = n^r$ (Example 13 of the unit proves this by FPM). (2) **Things not all different:** if among n things p_1 are alike of one kind, p_2 alike of another, ..., p_k alike of the k th kind, the number of arrangements of all n is $n!/(p_1! p_2! \dots p_k!)$. We divide because swapping identical things among themselves gives nothing new.

മലയാളത്തിൽ

ഒരേ അക്ഷരം പലതവണ ഉപയോഗിക്കാമെങ്കിൽ ഓരോ സ്ഥാനത്തിനും മുഴുവൻ n options കിട്ടും, അതുകൊണ്ട് n^r . വാക്കിൽ തന്നെ ഒരേ അക്ഷരം ആവർത്തിച്ചാൽ (BANANA-യിലെ മൂന്ന് A പോലെ), ആ A-കളെ പരസ്പരം മാറ്റിവെച്ചാൽ പുതിയ വാക്ക് ഉണ്ടാകില്ല; അതുകൊണ്ട് $n!$ -നെ ആ ആവർത്തനങ്ങളുടെ factorial കൊണ്ട് ഹരിക്കുന്നു. ഒരു പാഴ്ലറ്റിലെ ഒരേ നിറമുള്ള 3 ലഡ്ഡു പരസ്പരം മാറ്റിവെച്ചാൽ ആരും വ്യത്യാസം അറിയില്ല, അതുപോലെ.

r places, n things, repetition allowed: n^r

n things with $p_1, p_2, \dots, p_k$ alike ($p_1 + p_2 + \dots + p_k = n$): $n!/(p_1! p_2! \dots p_k!)$

EXAMPLE 12

How many different words can be formed using all the letters of BANANA?

6 letters: A three times, N twice, B once. Number = $6!/(3! 2!) = 720/12 = 60$.

EXAMPLE: MALAYALAM AND KERALA

How many arrangements of all the letters of (a) KERALA (b) MALAYALAM?

(a) 6 letters, A twice: $6!/2! = 360$.

(b) 9 letters: $A \times 4, M \times 2, L \times 2, Y \times 1$. Number = $9!/(4! 2! 2!) = 362880/96 = 3780$.

EXAMPLE 14

In how many ways can 5 letters be posted in 3 letter boxes?

Each letter has 3 choices of box, independently (a box can take many letters). Total = $3^5 = 243$.

Careful: it is not 5^3 . Ask "who is choosing?". Each letter chooses a box, so (boxes)^{letters}.

EXAMPLE: FOUR-LETTER WORDS

How many 4-letter words can be formed from a, b, g, h, k if (i) repetition is not allowed (ii) repetition is allowed?

(i) $5 \times 4 \times 3 \times 2 = {}^5P_4 = 120$

(ii) $5 \times 5 \times 5 \times 5 = 5^4 = 625$

EXAMPLE: SIGNALS WITH ALIKE FLAGS

How many signals can be made with 3 red, 4 white and 2 green flags, using all of them in a row?

9 flags, alike in groups of 3, 4, 2: $9!/(3! 4! 2!) = 362880/288 = 1260$.

EXAMPLE: LOTTERY TICKET NUMBERS

Suppose a lottery prints tickets with a series code of two different letters chosen from 12 letters, followed by a 6-digit number from 000000 to 999999. How many tickets can it print?

Series: $12 \times 11 = 132$ (no repetition). Number: each of 6 digits has 10 choices, 10^6 (repetition allowed).

Total = $132 \times 10^6 = 13,20,00,000$.

Circular permutation

Definition: A circular permutation is an arrangement of things around a circle; n different things can be arranged in a circle in $(n - 1)!$ ways, or $(n - 1)!/2$ if clockwise and anticlockwise orders are the same.

Circular permutation എന്നാൽ ഒരു വൃത്തത്തിന് ചുറ്റുമുള്ള വസ്തുക്കളുടെ arrangement; n വ്യത്യസ്ത വസ്തുക്കളെ വൃത്തത്തിൽ $(n - 1)!$ വിധത്തിൽ ക്രമീകരിക്കാം, clockwise-ഉം anticlockwise-ഉം ഒന്നായി കണക്കാക്കിയാൽ $(n - 1)!/2$ വിധത്തിൽ.

In a circle there is no first or last seat, so rotating everyone one place gives the same arrangement. ABCD, BCDA, CDAB, DABC are 4 line arrangements but one circular arrangement. So $n!$ line arrangements give $n!/n = (n - 1)!$ circular arrangements. Fix one person and arrange the rest. If clockwise and anticlockwise orders are also treated as the same (garlands, necklaces, beads, which can be turned over), divide again by 2.

മലയാളത്തിൽ

വട്ടമേശയ്ക്ക് ചുറ്റും ഇരിക്കുമ്പോൾ "ഒന്നാം സീറ്റ്" ഇല്ല; എല്ലാവരും ഒരു സീറ്റ് നീങ്ങിയാലും ആരുടെ അടുത്ത് ആര് എന്നത് മാറുന്നില്ല. അതുകൊണ്ട് ഒരാളെ ഉറപ്പിച്ചിരുത്തി ബാക്കി $(n - 1)$ പേരെ ക്രമീകരിക്കുക: $(n - 1)!$. തിരുവാതിരക്കളിയിൽ 6 പേർ വട്ടത്തിൽ നിൽക്കാൻ $5! = 120$ വിധം. പക്ഷേ മുല്ലപ്പൂമാല (garland) തിരിച്ചിട്ടാൽ clockwise, anticlockwise ഒന്നതന്നെ, അതുകൊണ്ട് $(n - 1)!/2$.

n different things in a circle (clockwise $\neq$ anticlockwise): $(n - 1)!$

Garland / necklace (clockwise = anticlockwise): $(n - 1)!/2$

Why: $n!$ line arrangements $\div n$ rotations = $(n - 1)!$

EXAMPLE 15

In how many ways can 10 students be arranged (i) in a line (ii) in a circle?

(i) $10! = 36,28,800$

(ii) $(10 - 1)! = 9! = 3,62,880$

EXAMPLE: GARLAND

How many different garlands can be made with 8 different flowers?

$(8 - 1)!/2 = 5040/2 = 2520$.

EXAMPLE: TWO PEOPLE ALWAYS TOGETHER

20 members of a college management sit at a round table. In how many ways, if the president (P) and vice president (V) always sit together?

Tie P and V into one unit. Now there are 19 units around the table: $(19 - 1)! = 18!$ ways. Inside the unit, P and V can sit as PV or VP: $2!$ ways.

Total = $2 \times 18!$.

EXAMPLE: ONAM SADHYA AT A ROUND TABLE

6 friends sit at a round table for sadhya. In how many ways can they sit so that two particular friends, Anu and Binu, are not next to each other?

All circular arrangements: $5! = 120$. Anu and Binu together: treat as one unit, 5 units in a circle $(5 - 1)! = 24$, times 2 for their order = 48.

Not together = $120 - 48 = 72$.

Exam tip: Read the object. People around a table: $(n - 1)!$. Beads, flowers, keys on a ring: $(n - 1)!/2$. For "always together", glue them into one unit and multiply by their internal arrangements; for "never together", subtract the "together" count from the total.

Combination: nC_r

Combination: a selection of things in which order does not matter.

nC_r : the number of combinations of n different things taken r at a time: ${}^nC_r = n!/(r!(n - r)!)$, $0 \leq r \leq n$.

Combination: ക്രമം പ്രധാനമല്ലാത്ത വസ്തുക്കളുടെ ഒരു selection.

nC_r : n വ്യത്യസ്ത വസ്തുക്കളിൽ നിന്ന് ഒരേ സമയം r എണ്ണം എടുത്തുള്ള combinations-ന്റെ എണ്ണം: ${}^nC_r = n!/(r!(n - r)!)$, $0 \leq r \leq n$.

A **combination** is a selection (a group), where order does not matter: the group $\{R, G\}$ is the same as $\{G, R\}$. The number of combinations of n different things taken r at a time ($0 \leq r \leq n$) is written nC_r , $C(n, r)$ or $(n \ r)$ in a column. Each group of r things can be arranged in $r!$ ways, so ${}^nP_r = {}^nC_r \times r!$, which gives the formula below.

മലയാളത്തിൽ

Combination എന്നാൽ വെറും തിരഞ്ഞെടുപ്പ്; ക്രമം നോക്കുന്നില്ല. 15 കളിക്കാരിൽ നിന്ന് 11 പേരുടെ ഒരു cricket team തിരഞ്ഞെടുക്കുമ്പോൾ ആരെ ആദ്യം വിളിച്ചു എന്നത് പ്രശ്നമല്ല, team ആരൊക്കെ എന്നത് മാത്രം. അതുകൊണ്ട് ${}^{15}C_{11} = 1365$ teams. Permutation-ൽ ഓരോ group-ഉം $r!$ വിധത്തിൽ ക്രമീകരിക്കും; ആ $r!$ കൊണ്ട് ഹരിച്ചാൽ combination കിട്ടും.

$${}^nC_r = n!/(r!(n - r)!) = {}^nP_r/r!, 0 \leq r \leq n$$

Quick form: ${}^nC_r = n(n - 1) \dots (n - r + 1)/r!$ e.g. ${}^7C_3 = (7 \cdot 6 \cdot 5)/(3 \cdot 2 \cdot 1) = 35$

Books R, G, W: 1 at a time R, G, W (3); 2 at a time RG, RW, GW (3); all 3: RGW (1)

EXAMPLE: ROUND-ROBIN MATCHES

Six cricket teams each play one match against every other team. How many matches?

A match is a pair of teams; India-vs-Pakistan is the same match as Pakistan-vs-India. So ${}^6C_2 = (6 \times 5)/2 = 15$.

EXAMPLE 18

A bag has 4 red and 7 white balls. In how many ways can 2 red and 3 white balls be drawn?

2 red from 4: ${}^4C_2 = 6$. 3 white from 7: ${}^7C_3 = 35$. Both are needed (AND): $6 \times 35 = 210$.

EXAMPLE: CRICKET TEAM FROM A SQUAD

The Kerala Ranji squad has 21 players: 11 batsmen, 7 bowlers, 3 wicket-keepers. In how many ways can a team of 11 with 6 batsmen, 4 bowlers and 1 wicket-keeper be chosen?

${}^{11}C_6 \times {}^7C_4 \times {}^3C_1 = 462 \times 35 \times 3 = 48510$.

EXAMPLE: AT LEAST

A committee of 5 is to be formed from 7 men and 5 women. In how many ways if it has (a) exactly 2 women (b) at least 3 women?

(a) 2 women AND 3 men: ${}^5C_2 \times {}^7C_3 = 10 \times 35 = 350$.

(b) 3W 2M OR 4W 1M OR 5W 0M:

${}^5C_3 \cdot {}^7C_2 + {}^5C_4 \cdot {}^7C_1 + {}^5C_5 \cdot {}^7C_0 = 10 \cdot 21 + 5 \cdot 7 + 1 \cdot 1 = 210 + 35 + 1 = 246$.

Exam tip: To compute nC_r fast, use the smaller of r and $n - r$: ${}^{18}C_{12} = {}^{18}C_6 = (18 \cdot 17 \cdot 16 \cdot 15 \cdot 14 \cdot 13)/720 = 18564$.

Permutation or combination? Properties of nC_r and problems

Definition: Use permutation (nP_r) when the order of the chosen things matters (arrangement) and combination (nC_r) when it does not (selection); ${}^nP_r = {}^nC_r \times r!$.

തിരഞ്ഞെടുത്ത വസ്തുക്കളുടെ ക്രമം പ്രധാനമാണെങ്കിൽ (arrangement) permutation (nP_r), പ്രധാനമല്ലെങ്കിൽ (selection) combination (nC_r) ഉപയോഗിക്കുക; ${}^nP_r = {}^nC_r \times r!$.

Use **permutation** when the order of the chosen things matters (arranging, ranking, seating, forming words or numbers, giving different posts or prizes). Use **combination** when only the group matters (selecting, choosing a team or committee, drawing cards or balls, shaking hands). The test: swap two of the chosen items. If you get a new outcome, it is a permutation; if not, it is a combination.

മലയാളത്തിൽ

സംശയം വന്നാൽ ഒരു ചോദ്യം ചോദിക്കുക: "തിരഞ്ഞെടുത്ത രണ്ടുപേരെ പരസ്പരം മാറ്റിയാൽ ഉത്തരം മാറുമോ?" 11 പേരുടെ cricket team-ൽ രണ്ടുപേരെ swap ചെയ്താൽ team അതുതന്നെ, അതുകൊണ്ട് combination. പക്ഷേ ആ 11-ൽ നിന്ന് captain, vice-captain എന്നിവരെ തിരഞ്ഞെടുത്താൽ, അവരെ swap ചെയ്യുമ്പോൾ പദവി മാറുന്നു, അതുകൊണ്ട് permutation. KL number plate-ൽ 12 എന്നതും 21 എന്നതും വേറെ നമ്പറുകൾ, അതും permutation.

Situation	Swap two items : new outcome?	Use	Count
Choose 11 players from 15	No	Combination	${}^{15}C_{11} = 1365$
Choose captain and vice-captain from 11	Yes (posts differ)	Permutation	${}^{11}P_2 = 110$
Seat 5 passengers in 8 bus seats	Yes	Permutation	${}^8P_5 = 6720$
Draw 1st, 2nd, 3rd prize tickets from 10	Yes (prizes differ)	Permutation	${}^{10}P_3 = 720$
Draw 3 equal consolation prizes from 10 tickets	No	Combination	${}^{10}C_3 = 120$
Handshakes among 10 people	No	Combination	${}^{10}C_2 = 45$
4-digit numbers from given digits	Yes	Permutation	depends on digits

Order matters (arrange, rank, word, number, seat, distinct posts) $\Rightarrow {}^nP_r$

Order does not matter (select, choose, group, team, committee, draw, handshake) $\Rightarrow {}^nC_r$

"Select then arrange" $\Rightarrow {}^nC_r \times r! = {}^nP_r$

Properties (Example 16):

(i) ${}^nC_r = {}^nC_{n-r}$ (ii) ${}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$ (Pascal's rule)

(iii) ${}^nC_0 = {}^nC_n = 1$ (iv) ${}^nC_1 = {}^nC_{n-1} = n$

If ${}^nC_x = {}^nC_y$ then $x = y$ or $x + y = n$

EXAMPLE 16: PROOFS

(i) ${}^nC_{n-r} = n!/((n-r)!(n-(n-r))!) = n!/((n-r)!r!) = {}^nC_r$. (Choosing r to take is the same as choosing $n-r$ to leave.)

$$\begin{aligned} \text{(ii)} \quad {}^nC_r + {}^nC_{r-1} &= n!/(r!(n-r)!) + n!/((r-1)!(n-r+1)!) \\ &= n!/((r-1)!(n-r)!) \times [1/r + 1/(n-r+1)] \\ &= n!/((r-1)!(n-r)!) \times (n-r+1+r)/(r(n-r+1)) \\ &= (n+1) \cdot n!/(r(n-r+1)(n-r)!) = (n+1)!/(r!(n+1-r)!) = {}^{n+1}C_r. \end{aligned}$$

(iii) ${}^nC_0 = n!/(0!n!) = 1$ and ${}^nC_n = n!/(n!0!) = 1$, using $0! = 1$.

(iv) ${}^nC_1 = n!/(1!(n-1)!) = n(n-1)!/(n-1)! = n$, and ${}^nC_{n-1} = n!/((n-1)!1!) = n$.

EXAMPLE 17: PLAYING CARDS

Four cards are chosen from a pack of 52 (4 suits of 13). In how many ways are they (i) of different suits (ii) of the same suit (iii) all face cards (iv) all red or all black?

(i) One from each suit: ${}^{13}C_1 \times {}^{13}C_1 \times {}^{13}C_1 \times {}^{13}C_1 = 13^4 = \mathbf{28561}$

(ii) All four from spades OR hearts OR diamonds OR clubs: $4 \times {}^{13}C_4 = 4 \times 715 = \mathbf{2860}$

(iii) 12 face cards (J, Q, K of each suit): ${}^{12}C_4 = \mathbf{495}$

(iv) 26 red OR 26 black: $2 \times {}^{26}C_4 = 2 \times 14950 = \mathbf{29900}$

EXAMPLE: SELECT, THEN ARRANGE

From a squad of 15, a team of 11 is chosen, and then a captain and vice-captain are named from the team. In how many ways?

Select team (combination): ${}^{15}C_{11} = 1365$. Then pick captain and vice-captain (permutation): ${}^{11}P_2 = 110$.

Both steps: $1365 \times 110 = \mathbf{1,50,150}$.

EXAMPLE: USING THE PROPERTIES

(a) If ${}^nC_8 = {}^nC_6$, find nC_2 . (b) If ${}^nC_2 = 45$, find n . (c) How many diagonals does an octagon have?

(a) $8 \neq 6$, so $8 + 6 = n$, $n = 14$. ${}^{14}C_2 = (14 \times 13)/2 = \mathbf{91}$.

(b) $n(n-1)/2 = 45 \Rightarrow n^2 - n - 90 = 0 \Rightarrow (n-10)(n+9) = 0 \Rightarrow \mathbf{n = 10}$.

(c) Any 2 of the 8 vertices give a line: ${}^8C_2 = 28$; 8 of these are sides. Diagonals = $28 - 8 = \mathbf{20}$.

Exam tip: Prizes: "3 different prizes to 5 students, no one gets more than one" is ${}^5P_3 = 60$ (prizes are different). If the prizes were identical it would be ${}^5C_3 = 10$. Read whether the objects are alike or different before choosing P or C.

Some important results: particular things included or excluded

s particular things always included: combinations: ${}^{n-s}C_{r-s}$; permutations: ${}^{n-s}P_{r-s} \times {}^rP_s$.

s particular things always excluded: combinations: ${}^{n-s}C_r$; permutations: ${}^{n-s}P_r$.

s particular things always included: combinations: ${}^{n-s}C_{r-s}$; permutations: ${}^{n-s}P_{r-s} \times {}^rP_s$.

s particular things always excluded: combinations: ${}^{n-s}C_r$; permutations: ${}^{n-s}P_r$.

Often some particular things must always be in (or never be in) the selection. For a **combination**: if s particular things are always included, they are already chosen, so choose the remaining r - s from the remaining n - s. If s things are always excluded, throw them out and choose all r from n - s. For a **permutation** with s things included: choose the other r - s things and arrange them (${}^{n-s}P_{r-s}$), then place the s special things into the r positions (rP_s).

മലയാളത്തിൽ

Captain-ഉം vice-captain-ഉം team-ൽ ഉറപ്പായും വേണം എന്ന് പറഞ്ഞാൽ അവർ രണ്ടുപേരും നേരത്തെ തന്നെ team-ൽ ഉണ്ട്; ബാക്കി സീറ്റുകൾ മാത്രം

ബാക്കിയുള്ളവരിൽ നിന്ന് നിറയ്ക്കുക. പരിശോധിക്കുക ഒരു കളിക്കാരനെ ഒഴിവാക്കണം എങ്കിൽ അയാളെ list-ൽ നിന്ന് വെട്ടിക്കളഞ്ഞ് ബാക്കിയുള്ളവരിൽ നിന്ന് മുഴുവൻ team-ഉം തിരഞ്ഞെടുക്കുക. Permutation ആണെങ്കിൽ ആ പ്രത്യേക ആളുകൾ ഏത് സ്ഥാനത്ത് നിൽക്കുന്നു എന്നതിനും (rP_s) എണ്ണം കൂട്ടണം.

Result I (permutations of n things, r at a time):

s particular things always included: ${}^{n-s}P_{r-s} \times {}^rP_s$

s particular things always excluded: ${}^{n-s}P_r$

Result II (combinations of n things, r at a time):

s particular things always included: ${}^{n-s}C_{r-s}$

s particular things always excluded: ${}^{n-s}C_r$

EXAMPLE 19

15 players are to be selected from 21. In how many ways if (i) 3 particular players are always included (ii) 2 particular players are always excluded?

(i) $n = 21, r = 15, s = 3$: ${}^{18}C_{12} = {}^{18}C_6 = (18 \cdot 17 \cdot 16 \cdot 15 \cdot 14 \cdot 13) / 720 = \mathbf{18564}$.

(ii) $n = 21, r = 15, s = 2$: ${}^{19}C_{15} = {}^{19}C_4 = (19 \cdot 18 \cdot 17 \cdot 16) / 24 = \mathbf{3876}$.

EXAMPLE 20

How many 5-letter words can be formed from the 8 letters a, b, c, d, e, f, g, h (no repetition) if (i) a and b are always included (ii) a, c, d are always excluded?

Words are arrangements, so use permutations.

(i) $n = 8, r = 5, s = 2$: ${}^6P_3 \times {}^5P_2 = 120 \times 20 = \mathbf{2400}$.

Check another way: choose the other 3 letters, ${}^6C_3 = 20$, then arrange all 5 letters, $5! = 120$: $20 \times 120 = 2400$.

(ii) $n = 8, r = 5, s = 3$: ${}^5P_5 = 5! = \mathbf{120}$.

EXAMPLE: CAPTAIN IN, INJURED PLAYER OUT

11 players are selected from 15. In how many ways if (i) captain and vice-captain are always included (ii) one injured player is always excluded?

(i) ${}^{13}C_9 = {}^{13}C_4 = (13 \cdot 12 \cdot 11 \cdot 10)/24 = \mathbf{715}$

(ii) ${}^{14}C_{11} = {}^{14}C_3 = (14 \cdot 13 \cdot 12)/6 = \mathbf{364}$

EXAMPLE: 4-DIGIT NUMBERS

How many 4-digit numbers (no repeated digit) can be formed from 1, 2, ..., 9 if (i) 1, 6, 8 are always included (ii) 3 and 8 are always excluded?

(i) $n = 9, r = 4, s = 3$: ${}^6P_1 \times {}^4P_3 = 6 \times 24 = \mathbf{144}$

(ii) $n = 9, r = 4, s = 2$: ${}^7P_4 = 7 \times 6 \times 5 \times 4 = \mathbf{840}$

Exam tip: First decide P or C, then write n, r, s explicitly, then use the formula. The most common error is using ${}^{n-s}P_{r-s}$ alone for "included" in permutations and forgetting the factor rP_s for placing the special items.

Binomial theorem for positive integral index

Definition: For a positive integer n, $(a + b)^n = \sum_{r=0}^n {}^nC_r a^{n-r} b^r$; the numbers nC_r are the binomial coefficients and $T_{r+1} = {}^nC_r a^{n-r} b^r$ is the general term.

Positive integer n, $(a + b)^n = \sum_{r=0}^n {}^nC_r a^{n-r} b^r$; nC_r are binomial coefficients, $T_{r+1} = {}^nC_r a^{n-r} b^r$ general term.

An expression with one term is a monomial ($7, 3x^2, x^2y$), with two terms a binomial ($a + b, x - 4y$), with three a trinomial ($a + b + c$), and with more a multinomial. The binomial theorem gives $(a + b)^n$ for $n = 1, 2, 3, \dots$ without repeated multiplication. The coefficients are the combinations ${}^nC_0, {}^nC_1, \dots, {}^nC_n$, called **binomial coefficients**: the term $a^{n-r} b^r$ appears nC_r times, because it comes from choosing b from r of the n brackets.

മലയാളത്തിൽ

സ്കൂളിൽ $(a + b)^2$, $(a + b)^3$ എന്നിവ മനഃപാഠം ആക്കിയിരുന്നു. ഇനി ഏത് n -നും ഒരൊറ്റ formula മതി. $(a + b)^n$ തവണ ഗുണിക്കുമ്പോൾ ഓരോ bracket-ൽ നിന്നും a അല്ലെങ്കിൽ b എടുക്കുന്നു; n -ൽ r bracket-ൽ നിന്ന് b എടുക്കാനുള്ള വഴികൾ nC_r . അതാണ് coefficient. a -യുടെ power n -ൽ നിന്ന് ഓരോ term-ലും ഒന്ന് കുറയും, b -യുടെ power 0-ൽ നിന്ന് ഒന്ന് കൂടും; രണ്ടും കൂട്ടിയാൽ എപ്പോഴും n .

$$(a + b)^n = {}^nC_0 a^n + {}^nC_1 a^{n-1}b + {}^nC_2 a^{n-2}b^2 + \dots + {}^nC_{n-1} ab^{n-1} + {}^nC_n b^n$$

$$\text{General } (r + 1)\text{th term: } T_{r+1} = {}^nC_r a^{n-r} b^r$$

Number of terms = $n + 1$; power of a + power of b = n in every term

For $(a - b)^n$ put $-b$: signs alternate $+$, $-$, $+$, $-$, $+$, $-$, ...

$$\text{Sum of coefficients: } {}^nC_0 + {}^nC_1 + \dots + {}^nC_n = 2^n \text{ (put } a = b = 1\text{)}$$

Pascal's triangle lists the coefficients; each entry is the sum of the two above it (this is ${}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$):

n = 1:			1	1			
n = 2:			1	2	1		
n = 3:			1	3	3	1	
n = 4:			1	4	6	4	1
n = 5:		1	5	10	10	5	1
n = 6:	1	6	15	20	15	6	1

EXAMPLE 21

Expand $(x + 1/x)^6$.

$a = x$, $b = 1/x$, $n = 6$; coefficients 1, 6, 15, 20, 15, 6, 1.

$$\begin{aligned} (x + 1/x)^6 &= x^6 + 6x^5(1/x) + 15x^4(1/x^2) + 20x^3(1/x^3) + 15x^2(1/x^4) + 6x(1/x^5) + 1/x^6 \\ &= x^6 + 6x^4 + 15x^2 + 20 + 15/x^2 + 6/x^4 + 1/x^6 \end{aligned}$$

EXAMPLE 22

Expand $(1 - x)^{10}$.

$a = 1$, $b = -x$, $n = 10$. Coefficients ${}^{10}C_r$: 1, 10, 45, 120, 210, 252, 210, 120, 45, 10, 1; signs alternate.

$$(1 - x)^{10} = 1 - 10x + 45x^2 - 120x^3 + 210x^4 - 252x^5 + 210x^6 - 120x^7 + 45x^8 - 10x^9 + x^{10}$$

EXAMPLE: $(1 + \sqrt{2})^5$

$a = 1$, $b = \sqrt{2}$, $n = 5$; coefficients 1, 5, 10, 10, 5, 1. Powers of $\sqrt{2}$: 1, $\sqrt{2}$, 2, $2\sqrt{2}$, 4, $4\sqrt{2}$.

$$= 1 + 5\sqrt{2} + 10(2) + 10(2\sqrt{2}) + 5(4) + 4\sqrt{2} = (1 + 20 + 20) + (5 + 20 + 4)\sqrt{2} = 41 + 29\sqrt{2}.$$

EXAMPLE: $(3x - y)^7$

$a = 3x, b = -y$, coefficients 1, 7, 21, 35, 35, 21, 7, 1; powers of 3: 2187, 729, 243, 81, 27, 9, 3, 1.
 $= 2187x^7 - 7(729)x^6y + 21(243)x^5y^2 - 35(81)x^4y^3 + 35(27)x^3y^4 - 21(9)x^2y^5 + 7(3)xy^6 - y^7$
 $= 2187x^7 - 5103x^6y + 5103x^5y^2 - 2835x^4y^3 + 945x^3y^4 - 189x^2y^5 + 21xy^6 - y^7$

EXAMPLE: PARTICULAR TERMS USING T_{R+1}

(a) Coefficient of x^5 in $(x + 2)^8$. (b) Middle term of $(2x - 3)^6$. (c) Term independent of x in $(x^2 + 1/x)^9$.

(a) $T_{r+1} = {}^8C_r x^{8-r} 2^r$. Need $8 - r = 5$, so $r = 3$: ${}^8C_3 \times 2^3 = 56 \times 8 = 448$.

(b) $n = 6$, so 7 terms; the middle one is T_4 ($r = 3$): ${}^6C_3 (2x)^3 (-3)^3 = 20 \times 8x^3 \times (-27) = -4320x^3$.

(c) $T_{r+1} = {}^9C_r (x^2)^{9-r} (1/x)^r = {}^9C_r x^{18-3r}$. Need $18 - 3r = 0$, $r = 6$: ${}^9C_6 = 84$.

Exam tip: For $(a - b)^n$ keep the minus sign inside $b = -y$ and let the powers decide the sign; odd powers are negative. When b is a constant like $3x$ or 2 , raise the whole thing to the power: $(3x)^4 = 81x^4$, not $3x^4$.

Binomial theorem for rational index

Definition: For any rational n and $|x| < 1$, $(1 + x)^n = 1 + nx + n(n - 1)x^2/2! + \dots + n(n - 1)\dots(n - r + 1)x^r/r! + \dots$, an infinite series unless n is a non-negative integer.

എത് rational n -നും $|x| < 1$ ആകുമ്പോൾ, $(1 + x)^n = 1 + nx + n(n - 1)x^2/2! + \dots + n(n - 1)\dots(n - r + 1)x^r/r! + \dots$; n ഒരു non-negative integer അല്ലെങ്കിൽ ഇത് ഒരു infinite series ആണ്.

Writing nC_r as $n(n - 1)\dots(n - r + 1)/r!$ lets the formula work when n is a negative integer or a fraction. The series then never stops (it is infinite), and it gives a finite sum only when the second term is small: for $(1 + x)^n$ we need $|x| < 1$. For $(a + b)^n$, take out the bigger term first: $a^n(1 + b/a)^n$ if $|b/a| < 1$, or $b^n(1 + a/b)^n$ if $|a/b| < 1$.

മലയാളത്തിൽ

n ഒരു positive integer അല്ലെങ്കിൽ (ഉദാ: $-4, 1/2, 2/3$) expansion അവസാനിക്കില്ല; terms അനന്തമായി തുടരും. അപ്പോൾ x ചെറുതാണെങ്കിൽ ($|x| < 1$) മാത്രമേ ഈ അനന്ത ശ്രേണിക്ക് ഒരു finite value ഉണ്ടാകൂ. അതുകൊണ്ട് ആദ്യം bracket-നെ $(1 + \text{ചെറിയ സംഖ്യ})$ എന്ന രൂപത്തിൽ ആക്കുക, ഉദാഹരണം $(5 + 3x) = 5(1 + 3x/5)$. ഉത്തരത്തോടൊപ്പം "valid if $|x| < \dots$ " എന്ന condition എഴുതാൻ മറക്കരുത്; അതിന് മാർക്കുണ്ട്.

$(1+x)^n = 1 + nx + n(n-1)x^2/2! + n(n-1)(n-2)x^3/3! + \dots + n(n-1)\dots(n-r+1)x^r/r! + \dots, \quad |x| < 1$
 $(a+b)^n = a^n(1+b/a)^n$ if $|b/a| < 1$; $= b^n(1+a/b)^n$ if $|a/b| < 1$
 Useful: $(1-x)^{-1} = 1 + x + x^2 + \dots$, $(1+x)^{-1} = 1 - x + x^2 - \dots$, $(1+x)^{-2} = 1 - 2x + 3x^2 - 4x^3 + \dots$

EXAMPLE 23

Expand $(5 + 3x)^{-4}$.

$$\begin{aligned}
 (5 + 3x)^{-4} &= 5^{-4}(1 + 3x/5)^{-4}. \text{ Here } n = -4, \text{ and } x \text{ is replaced by } 3x/5. \\
 &= (1/625)[1 + (-4)(3x/5) + ((-4)(-5)/2!)(3x/5)^2 + ((-4)(-5)(-6)/3!)(3x/5)^3 + \dots] \\
 &= (1/625)[1 - 12x/5 + 10(9x^2/25) - 20(27x^3/125) + \dots] \\
 &= (1/625)[1 - (12/5)x + (18/5)x^2 - (108/25)x^3 + \dots]
 \end{aligned}$$

Valid when $|3x/5| < 1$, i.e. $|x| < 5/3$.

EXAMPLE 24

Expand $(7 - 2x)^{2/3}$.

$$\begin{aligned}
 &= 7^{2/3}(1 - 2x/7)^{2/3}, n = 2/3, "x" = -2x/7. \\
 &= 7^{2/3}[1 + (2/3)(-2x/7) + ((2/3)(-1/3)/2!)(-2x/7)^2 + \dots] \\
 &= 7^{2/3}[1 - 4x/21 - (1/9)(4x^2/49) + \dots] = 7^{2/3}[1 - (4/21)x - (4/441)x^2 - \dots]
 \end{aligned}$$

Valid when $|2x/7| < 1$, i.e. $|x| < 7/2$.

EXAMPLE: $(1 - 3x)^{1/2}$

$$\begin{aligned}
 n &= 1/2, "x" = -3x. \\
 &= 1 + (1/2)(-3x) + ((1/2)(-1/2)/2!)(9x^2) + ((1/2)(-1/2)(-3/2)/3!)(-27x^3) + \dots \\
 &= 1 - (3/2)x + (-1/8)(9x^2) + (1/16)(-27x^3) + \dots \\
 &= 1 - (3/2)x - (9/8)x^2 - (27/16)x^3 - \dots, \text{ valid when } |3x| < 1, \text{ i.e. } |x| < 1/3.
 \end{aligned}$$

EXAMPLE: APPROXIMATION

Find $\sqrt{0.98}$ to 5 decimal places.

$$\sqrt{0.98} = (1 - 0.02)^{1/2} = 1 + (1/2)(-0.02) + ((1/2)(-1/2)/2)(0.0004) + \dots = 1 - 0.01 - 0.00005 - \dots \approx \mathbf{0.98995}.$$

The next term is about -0.0000005 , so it does not affect the fifth decimal place.

Exam tip: The two usual mistakes are (1) not taking a^n outside the bracket before expanding, and (2) squaring or cubing only x and forgetting the coefficient: $(-3x)^3 = -27x^3$. Always end with the validity condition on $|x|$.

Quick recap

- $n! = n(n-1)\dots 2 \cdot 1$, $0! = 1$; open the larger factorial to cancel: $n! = n(n-1)!$.

- FPM (AND, in succession): multiply the number of ways. FPA (OR, mutually exclusive): add.
- Permutation = arrangement (order matters): ${}^nP_r = n!/(n-r)!$.
- Repetition allowed: n^r . Alike things: $n!/(p_1! p_2! \dots p_k!)$.
- Circular: $(n-1)!$; garland or necklace: $(n-1)!/2$.
- "Always together": glue into one unit, multiply by their internal arrangements. "Never together" = total – together.
- Combination = selection (order does not matter): ${}^nC_r = n!/(r!(n-r)!) = {}^nP_r/r!$.
- Decision test: swap two chosen items; new outcome $\Rightarrow$ permutation, same outcome $\Rightarrow$ combination.
- ${}^nC_r = {}^nC_{n-r}$; ${}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$; ${}^nC_0 = {}^nC_n = 1$; ${}^nC_1 = n$.
- s things included: $C \Rightarrow {}^{n-s}C_{r-s}$, $P \Rightarrow {}^{n-s}P_{r-s} \times {}^rP_s$. Excluded: ${}^{n-s}C_r$, ${}^{n-s}P_r$.
- $(a+b)^n = \sum {}^nC_r a^{n-r} b^r$ ($r = 0$ to n); $n+1$ terms; $T_{r+1} = {}^nC_r a^{n-r} b^r$.
- Rational index: $(1+x)^n = 1 + nx + n(n-1)x^2/2! + \dots$, infinite series, valid only for $|x| < 1$.

Check yourself

Q. 1. Evaluate $22!/19!$ and $15!/(10! 5!)$. Then solve $n! = 72(n-2)!$.

$$22!/19! = 22 \times 21 \times 20 = 9240.$$

$$15!/(10! 5!) = (15 \times 14 \times 13 \times 12 \times 11)/120 = 3003.$$

$$n(n-1)(n-2)! = 72(n-2)! \Rightarrow n^2 - n - 72 = 0 \Rightarrow (n-9)(n+8) = 0 \Rightarrow n = 9 \text{ (n cannot be negative).}$$

Q. 2. Solve $(n-2)! = 12(n-4)!$, and express $3 \cdot 6 \cdot 9 \cdot 12 \cdot 15$ in factorials.

$$(n-2)(n-3)(n-4)! = 12(n-4)! \Rightarrow n^2 - 5n - 6 = 0 \Rightarrow (n-6)(n+1) = 0 \Rightarrow n = 6.$$

$$3 \cdot 6 \cdot 9 \cdot 12 \cdot 15 = 3^5(1 \cdot 2 \cdot 3 \cdot 4 \cdot 5) = 3^5 \cdot 5!.$$

Q. 3. An exam has 10 multiple-choice questions: the first five have 4 options each and the last five have 5 options each. How many answer sequences are possible?

$$\text{Each question is answered (AND), so by FPM: } 4^5 \times 5^5 = 1024 \times 3125 = 32,00,000.$$

Q. 4. Find n if ${}^5P_n = {}^6P_{n-1}$.

$$5!/(5-n)! = 6!/(7-n)! \Rightarrow (7-n)!/(5-n)! = 6 \Rightarrow (7-n)(6-n) = 6 \Rightarrow n^2 - 13n + 36 = 0 \Rightarrow (n-4)(n-9) = 0.$$

$$n = 9 \text{ is impossible since } {}^5P_9 \text{ has no meaning (cannot arrange 9 of 5 things). So } n = 4. \text{ Check: } {}^5P_4 = 120 = {}^6P_3.$$

Q. 5. In how many ways can 3 different prizes be given to 5 students if (i) no student gets more than one prize (ii) a student may get any number of prizes (iii) no student gets all the prizes?

(i) $5 \times 4 \times 3 = {}^5P_3 = 60$.

(ii) Each prize has 5 choices: $5^3 = 125$.

(iii) Remove the 5 cases where one student takes all three: $125 - 5 = 120$.

Q. 6. How many words can be formed from the letters of AMAR? How many garlands can be made from 8 different flowers?

AMAR: 4 letters with A twice: $4!/2! = 12$.

Garland: $(8 - 1)!/2 = 5040/2 = 2520$.

Q. 7. From 21 cricketers (11 batsmen, 7 bowlers, 3 wicket-keepers), in how many ways can 11 be picked with 6 batsmen, 4 bowlers and 1 wicket-keeper?

Selection, so combinations; all three parts needed, so multiply: ${}^{11}C_6 \times {}^7C_4 \times {}^3C_1 = 462 \times 35 \times 3 = 48510$.

Q. 8. How many 4-digit numbers (no repeated digit) can be formed from 1–9 if (i) 1, 6, 8 are always included (ii) 3 and 8 are always excluded?

Numbers are arrangements $\Rightarrow$ Result I.

(i) $n = 9, r = 4, s = 3: {}^6P_1 \times {}^4P_3 = 6 \times 24 = 144$.

(ii) $n = 9, r = 4, s = 2: {}^7P_4 = 840$.

Q. 9. Find the term independent of x in $(x + 1/x)^6$ and the coefficient of x^4y^3 in $(3x - y)^7$.

$(x + 1/x)^6: T_{r+1} = {}^6C_r x^{6-2r}; 6 - 2r = 0 \Rightarrow r = 3; \text{term} = {}^6C_3 = 20$.

$(3x - y)^7: r = 3 \text{ gives } {}^7C_3 (3x)^4 (-y)^3 = 35 \times 81 \times (-1) x^4 y^3; \text{coefficient} = -2835$.

Q. 10. Expand $(1 + x)^{-2}$ up to x^3 and state when it is valid. Use it to write $1/(1.01)^2$ approximately.

$n = -2: 1 + (-2)x + ((-2)(-3)/2!)x^2 + ((-2)(-3)(-4)/3!)x^3 + \dots = 1 - 2x + 3x^2 - 4x^3 + \dots$, valid for $|x| < 1$.

$x = 0.01: 1 - 0.02 + 0.0003 - 0.000004 \approx 0.980296$.

Based on the IGNOU unit texts. Check definitions against the PDFs before the exam.